

Comparative Assessment of Accuracy in Video-based Monocular Human Pose Estimation Frameworks

Fabian Kahl, Philipp Wegner, Maximilian Kapsecker, Leon Nissen, Jennifer Faber, Stephan M. Jonas, Lara Marie Reimer

Abstract—In human pose estimation, a comprehensive evaluation of state-of-the-art frameworks is necessary to advance both research and practical applications. This paper presents a thorough review of state-of-the-art 2D and 3D human pose estimation frameworks, analyzing 118 papers and four GitHub repositories, with a focus on frameworks made since 2019. The following frameworks are chosen based on predefined inclusion criteria: lphaPose, Detectron2, MediaPipe, MeTR bs, MHFormer, MMPose, MoveNet, OpenPifPaf, OpenPifPaf-vita, OpenPose, PoseFormerV2, rtlib, StridedTransformer-Pose3D, ultralytics (YOLOv8), ViTPose, and YOLOv7. This paper evaluates these 16 frameworks on an existing, unpublished dataset consisting of exercise videos recorded with a monocular RGB camera and synchronized gold-standard motion capture data. The dataset includes videos of nine individuals performing eight exercises, recorded from two camera views with different planar angles. The analysis evaluates joint angle performance of the frameworks using weighted mean absolute error and weighted intraclass correlation coefficient as quantitative metrics. MeTR bs emerged as the best overall framework, while lphaPose, rtlib, and YOLOv7 were the top 2D performers. The used code is available open source¹.

Index Terms—Person Keypoint Detection, Pose Landmark Detection, Single-View, Machine Learning, Framework Comparison.

I. INTRODUCTION

THE objective of human pose estimation (HPE) involves determining the positions of body joints from sensor data, primarily images or videos [1].

The widespread adoption of deep learning-based architectures, like AlexNet [2], VGG [3], ResNet [4], and Inception [5], along with the availability of extensive datasets, such as ImageNet [6], Human3.6M [7], MPII Human Pose [8],

COCO [9], CMU Panoptic [10]–[12], and COCO-WholeBody [13], [14] has led to the development of a multitude of HPE solutions, as mentioned in several surveys [1], [15], [16].

Surveys conducted by Moeslund et al. (2006), Kale and Patil (2016), and Zheng et al. (2023) highlight the breadth of these applications [15], [17], [18]. In healthcare, HPE is used for medical diagnostics and rehabilitation. In sports, it can support posture correction, thereby optimizing athlete performance. Additionally, HPE finds applications in fields such as robotics and surveillance. HPE also plays a role in various sectors, including gaming, augmented and virtual realities, sign language recognition, device control, animation, and parsing clothing-related information.

Researchers have explored the use of multi-view setups, like optoelectronic measurement systems (OMSs), or additional sensors to measure depth, such as electromagnetic measurement systems, ultrasonic localization systems, and inertial sensor measurement systems, to generate and tackle a well-posed problem of measuring 3D coordinates of joints [15], [19]. However, these approaches are typically not cost-effective, and their complex setups limit their application to more or less professional settings [15], [19]. Notably, Kinect, a widely studied HPE system, uses an RGB camera and an infrared sensor for 3D motion capture, yet still requires a specialized setup [20], [21].

Nevertheless, OMSs are widely recognized as a gold standard for achieving high accuracy in motion capture [19]. These systems utilize triangulation through multiple synchronized cameras, enabling the reconstruction of three-dimensional positions [19]. Many research studies have adopted OMSs such as Vicon [22] and Qualisys [23] as benchmarks for evaluating the accuracy of other motion capture systems [21], including HPE methods [24]–[29].

To overcome the requirement of a specialized setup, researchers have simplified the input to consumer-grade monocular 2D images or videos while extrapolating the output to 3D, as shown by Sárárdi et al. (2021) and Li et al. (2022) [30], [31]. While this approach offers greater flexibility in open environments, it creates an ill-posed problem due to the intrinsic limitations of extracting depth information from two-dimensional visual data. Consequently, this can lead to uncertain estimations, especially in complex scenarios.

To overcome the ill-posed problem, Cao et al. (2019) and

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¹<https://github.com/digital-medicine/HPEFCComparison>

Fang et al. (2022) reduced the output dimensionality to 2D [32], [33]. This effectively transformed the problem of joint keypoint detection into a well-posed one. However, these methods provided insights only within the image or video plane, failing to capture the spatial relationships among body joints in the three-dimensional world.

Furthermore, several studies have demonstrated that various factors influence the reliability and validity of monocular HPE methods, including the recorded task, video quality, and camera position [34]–[36].

To the best of our knowledge, no comprehensive study has compared the performance of more than five frameworks using monocular video input data (see Section III). This study presents an analysis of existing frameworks that implement 2D and 3D HPE methods on a yet-unpublished dataset of monocular videos introduced by Reimer et al. (2022) [37]. Instead of utilizing well-known datasets such as COCO [9] or Human3.6M [7], the evaluation relies on the dataset introduced by Reimer et al. (2022) [37] (see Section IV-E for details), containing participants performing a range of exercises. This choice ensured that the models in the frameworks had no prior exposure to any video data. We calculated joint angles from coordinates originally captured with Vicon, a marker-based OMS, and used them as the reference system (see Section II-B). The analysis focused on addressing the following research questions (RQ):

- RQ 1: How accurate are the chosen frameworks compared to Vicon?
- RQ 2: Can 2D frameworks perform comparably to 3D frameworks in Side View scenarios where depth information is significantly limited?
- RQ 3: What are the main factors influencing HPE performance?

II. BACKGROUND

A. Human Pose Estimation Categories

We categorized the HPE methods into three main characteristics: the dimensionality of the HPE output (2D vs. 3D), the maximum number of people analyzed in the scene (single-person vs. multi-person), and the approach of capturing poses (top-down vs. bottom-up vs. two-stage approach).

1) *2D vs. 3D Output*: 2D methods focus on estimating the poses of people within a 2D space [33], [38]. In contrast, 3D methods aim to extrapolate the poses into 3D space [31], [39].

2) *Single vs. Multi-Person Pose Estimation*: Single-person methods focus on estimating a single person's pose [40]. Conversely, multi-person frameworks simultaneously assess the poses of multiple people present in the scene [32], [38], [41].

3) *Top-down vs. Bottom-up vs. Two-Stage approaches*: HPE methods employed two main strategies for estimating people's pose in 2D images or video frames: top-down and bottom-up approaches. Top-down approaches operate sequentially, progressing from lower to higher resolutions [33]. They break down the task into single-person detection problems by identifying person instances, for example, using a bounding box object detector [33]. Subsequently, they perform a

single-person pose estimation for each instance. This enables normalizing people's sizes to a similar scale by adjusting the detected person bounding boxes, making it less sensitive to scale variations [33]. However, they have limitations. Unsuccessful person detection, called the early commitment challenge, cannot be easily corrected [16], [42]. For example, this might happen if multiple people are in close proximity [16]. Additionally, the computational costs increase with the number of people in the image, resulting in a processing time that is directly proportional to the population size [16], [42].

Alternatively, bottom-up approaches adopt a reverse process from higher to lower resolutions. These approaches focus on localizing identity-free semantic entities, such as keypoints, within an image or video frame and subsequently grouping them into person instances [32]. By processing from high to low resolutions, bottom-up approaches overcome the early commitment challenge of top-down approaches. They demonstrate a detached runtime complexity from the number of people in the image, offering potential computational advantages [32]. However, bottom-up approaches face challenges in effectively grouping body parts when significant overlap exists between people [43]. Additionally, these approaches may struggle to address scale variation, leading to potential performance disparities compared to top-down methods [44], especially for scale variations of multiple people within a single image or video frame [45].

When it comes to 3D estimation, both approaches are also feasible. A third approach is the two-stage approach. This method divides the problem into two stages: first, it performs a 2D pose estimation using a well-established 2D HPE framework. Then, it lifts the estimated 2D poses to 3D using upsampling models [39], [46]–[48].

B. Gold Standard: Vicon

Vicon is a company specializing in motion capture technology and solutions [22]. They provide systems and software for capturing and analyzing human and object movements. The associated OMS, referred to here as Vicon, uses infrared cameras and markers to track and record the motion of subjects, enabling detailed analysis and visualization of movement patterns. The measurement accuracy of Vicon was validated by Merriault et al. (2017), revealing challenges in dynamic scenarios. Key issues included measurement errors that varied with speed and marker positioning, with higher speeds generally leading to reduced errors. System performance was also affected by factors such as camera calibration and marker placement, leading to variations in accuracy across different markers. Despite these challenges, the errors remained within a 2 mm range [49]. Due to its high accuracy, many research studies have adopted Vicon as a benchmark for evaluating the performance of HPE methods [24]–[26], [29].

III. RELATED WORK

Several studies have investigated the performance of different monocular HPE frameworks in various scenarios involving exercise videos.

Ota et al. (2020) compared OpenPose and Vicon using exercise videos of 20 healthy young people doing bilateral squats [24]. They captured the videos from the right side at 60 Hz. They focused on several joint angles. Both methods demonstrated high reliability, as indicated by the intraclass correlation coefficient (ICC), specifically ICC(1,3) (Vicon: 0.87-0.93, OpenPose: 0.92-0.96). OpenPose's validity was assessed against Vicon, showing varying ICC(2,1), and R^2 values (ICC(2,1): 0.37-0.82, R^2 : 0.26-0.83). The Bland-Altman analysis revealed biases. OpenPose underestimated trunk angles and overestimated hip angles, with fixed biases in knee and ankle angles.

To aid interpretation, we briefly explain the ICC notation used across several studies. Following the Shrout and Fleiss convention, ICC(a,b) specifies the model and measurement type. The first index a identifies the statistical model. ICC(1,b) is a one-way random-effects model, ICC(2,b) is a two-way random-effects model assessing absolute agreement, and ICC(3,b) is a two-way mixed-effects model assessing consistency. The second index b denotes whether the ICC is computed on a single measurement ($b = 1$) or on the mean of multiple measurements ($b > 1$). Higher ICC values represent greater agreement [50].

Similarly, Ota et al. (2021) examined OpenPose and Vicon in the context of treadmill walking and running with 24 healthy young participants [25]. They recorded videos from the right side and the back at 60 Hz. They focused on multiple parameters for several joints. Pelvic parameters exhibit low to moderate agreement (ICC(2,1): -0.06-0.23, R^2 : 0.00-0.20) with fixed and proportional biases in a Bland-Altman analysis. Hip measurements show variable agreement (ICC(2,1): 0.03-0.86, R^2 : 0.00-0.81) with similar biases. Knee measurements indicate moderate to high agreement (ICC(2,1): 0.23-0.93, R^2 : 0.41-0.92), primarily involving fixed biases. Ankle measurements show moderate agreement (ICC(2,1): 0.11-0.68, R^2 : 0.14-0.66), with mostly fixed biases.

VanKeersbilck et al. (2022) extended the analysis to include MediaPipe compared to Vicon [26]. This study focused on a single person running, with the video captured from the front at 240 Hz. They reported high correlation values for ankle, hip, and knee joint angles (MediaPipe: 0.928-0.983, OpenPose: 0.904-0.975), but with limitations in predicting maximum flexion and extension angles.

Hii et al. (2023) assessed the accuracy of MediaPipe compared to Vicon during gait exercises [29]. They included 32 healthy participants (22 males and 10 females) and protected participant privacy by blurring faces in videos. Results showed ICC(2,1) for temporal gait parameters ranged from 0.47 to 0.95 with mean absolute errors (MAEs) ranging from 30 to 50 ms. For mean values per person, they reported ICC(2,1) values ranging from 0.55 to 0.96 and MAEs ranging from 20 to 40 ms.

Lafayette et al. (2022) compared MediaPipe with Qualisys in repetitive limb movements [28]. Six participants raised and lowered their right limbs 120 times across ten recordings. They captured the videos from the front and at a 30° angle at 30 Hz. MediaPipe achieved an MAE of $8.57^\circ \pm 3.06^\circ$ with a Pearson correlation of 0.86.

Murugan et al. (2024) compared MediaPipe, VideoPose3D (they referred to it as VP3D_FB), MMPose (they referred to it as VP3D), 3D-pose-baseline, and PSTMO against a motion capturing system from Motion Analysis Corporation [51]. The evaluation used weighted MAEs across 10 joint angles, based on a dataset of 15 subjects performing eight exercises with objects, each lasting 2 minutes, captured from three camera angles at 30 Hz. The study focused on rankings from “a” to “c”, rather than providing detailed, quantitative results, with VideoPose3D emerging as the top-performing model.

Mohammed et al. (2023) compared MeTRs with Windi. This study observed eight joints for ten people performing three exercises [52]. Each participant performed each exercise 10 times, with videos taken from angles of 45° and frontal. They found discrepancies between the joint angles provided by Windi and calculated angles based on the Windi-provided skeletons. The angle deviations between Windi and MeTRs varied across joints and exercises, with median values ranging from 1.27° to 43.94°, mean values from 1.89° to 45.73°, and maximum errors from 7.70° to 73.02°.

Ceriola et al. (2023) compared MMPose, MediaPipe, and LphaPose against a blob-analysis algorithm, which they claim is less accurate than an optoelectronic system [53]. Their study focused on three one-minute videos capturing six anatomical landmarks of the right leg, showcasing one subject running on a treadmill at various speeds, with 60 Hz recorded from the side. They compared the root mean square error (RMSE) for the X and Y axes on three one-minute videos. MMPose yielded the most favorable outcomes among the tested frameworks, with a mean RMSE/mm for the X axis ranging from 4.5 to 35 and for the Y axis ranging from 4.7 to 16.5, depending on the speed and landmark.

Wang et al. (2021) conducted a comparison among OpenPose, Detectron2, and LphaPose on the publicly available COCO 2015 test images dataset² instead of videos [54]. Given the dataset's accessibility, it might be possible that some of the predefined models within the frameworks have been trained on it or parts of it, though the study does not provide explicit details on this. Therefore, some degree of data overlap between training and test data should be kept in mind when interpreting the results. The evaluation involved calculating various average precision (AP) values. LphaPose outperformed OpenPose and Detectron2 in all taken measurements, with an AP over intersection over union (IoU) thresholds from 0.5 to 0.95 of 73.3, an AP at an IoU threshold of 0.5 of 89.2, an AP at an IoU threshold of 0.75 of 79.1, an AP medium of 69.0, and an AP large of 78.6. Additionally, LphaPose exhibited the highest frame rate, at 20 frames per second.

Sabo et al. (2023) analyzed five frames featuring the most extreme joint angles from 97 videos [55]. These frames were selected via random uniform sampling of 20 frames per video, followed by manual selection of the frame in which the joint of interest was most hyperextended. They evaluated eight body angles using the HPE frameworks LphaPose, Detectron2, MediaPipe, MoveNet, and OpenPose, comparing the results with goniometer measurements and manual annotations. For

²<http://images.cocodataset.org/zips/test2015.zip>

two-hand angles, they assessed lphaPose and MediaPipe-Hands using the same comparison. The study found moderate correlations between the HPE frameworks and goniometer measurements for all angles, except the ankle. Similarly, manually annotated angles were moderately to strongly correlated with goniometer measurements for all but one angle.

Docekal et al. (2022) evaluated 2D whole-body keypoint detection on static images, comparing OpenPose, MMPose, lphaPose, and Detectron2 against Qualisys [27]. They assessed the median distance to the true value in centimeters and the number of keypoints with confidence above two thresholds.

lphaPose and MMPose emerged as the best-performing frameworks. Li et al. (2023) compared lphaPose, Detectron2, MediaPipe, OpenPose, and YOLOv7 in a single video of a person [56]. They focused on comparing detection details, possible keypoints, and usability. They did not compare the performance of the frameworks' estimations.

While these works provide valuable insights into the capabilities of different HPE frameworks, comparing them proved challenging. For instance, while some studies used ICC(2,1) to assess knee or elbow angles, others did not. Additionally, the angles analyzed varied: some studies focused solely on the lower body, some on the upper body, and some on both. To the best of our knowledge, no study has comprehensively compared more than five frameworks. Additionally, no study has compared more than five frameworks with a gold-standard motion-capturing tool for monocular videos or explored potential gender biases in this context.

IV. MATERIALS & METHODS

A. Study Design

We conducted a controlled experiment evaluating the performance of the frameworks introduced in Section IV-B. The analysis compared joint angles estimated by these frameworks with those measured by Vicon for nine participants performing eight exercises with ten repetitions each. Reimer et al. (2022) [37] provide the exact setup of the Vicon system. The dataset used in this experiment, also introduced by Reimer et al. (2022), is described in Section IV-E. Instead of relying on widely recognized datasets such as COCO [9] or Human3.6M [7], we decided to use this dataset to ensure that none of the frameworks had prior exposure to the data during training.

To quantify the disparities between the frameworks and Vicon, we computed weighted MSE and weighted ICC(2,1). Additionally, we conducted a factor analysis using Welch's ANOVA and Welch's *t*-tests to identify factors contributing to the variations between the frameworks and Vicon.

B. Framework Selection Criteria

This study includes a review of existing literature on HPE frameworks. We started our review with the survey by Zheng et al. (2023) [15] and the study by Li et al. (2023) [56]. Building on their work, the search expanded through additional screening of identified publications and repositories. The goal was to identify frameworks contributing to the current state of the art. To ensure the inclusion of up-to-date and accessible resources, the review considered only papers published in or

after 2019 that used monocular 2D input. Out of 118 papers, 41 lacked available code, leaving 77 eligible frameworks for analysis. In addition, the study includes four frameworks for which no corresponding academic papers were available. Each had more than 1,000 GitHub commits. We saw this as a proxy for sustained development activity of frameworks without formal publications. Two frameworks, OpenPifPaf and OpenPifPaf-vita [57], referred to the same publication.

After applying these initial screening criteria, $77+4+1 = 82$ frameworks (from papers + from GitHub + extra framework from a paper that already proposed a framework) were left. To refine the selection, the following framework inclusion criteria were used and independently evaluated by two reviewers in descending order. Missing one inclusion criterion led to not checking the further criteria. If the two reviewers had differing opinions on a framework, they jointly re-evaluated it until they reached a consensus. The criteria are:

- **Up-to-Date:** Frameworks had to receive recent code updates relevant to HPE capability, made on or after January 1, 2023, and before the date of screening (June 28, 2024), to ensure they were up-to-date. Updates to only the documentation files were not sufficient. (24 remaining)
- **Ease of Application:** Frameworks had to offer ease of use through a demonstration or user-friendly documentation for in-the-wild images or videos in understandable English. (20 remaining)
- **Video Processing Capability:** Frameworks had to be able to process video data. (18 remaining)
- **Availability of Models:** Pre-trained models had to be open-provision compatible with the provided code. (16 remaining)
- **Skeleton Information:** The models had to be able to provide limb and torso skeleton information. This is crucial for analyzing joint angles in human body models. (16 remaining)

Applying these criteria reduced the selection from 118 papers and four GitHub repositories to 16 frameworks (see Appendix Table I): lphaPose, Detectron2, MediaPipe, MeTRBS, MHFormer, MMPose, MoveNet, OpenPifPaf, OpenPifPaf-vita, OpenPose, PoseFormerV2, rtm-lib, StridedTransformer-Pose3D, ultralytics, ViTPose, and YOLOv7.

C. Human Pose Estimation Frameworks

This study compared 16 HPE frameworks, nine generating 2D output and seven generating 3D output. Among them, 11 frameworks could detect multiple individuals within a single frame, while five were designed solely for single-person pose estimation. Five frameworks used a top-down approach, six used a bottom-up approach, and five used a two-stage approach. Eight frameworks could provide some hand-keypoint information, 13 could provide some face-keypoint information, and seven could provide some foot-keypoint information. However, this study did not evaluate these additional parameters. Eight frameworks included a fully functional demo that saved keypoint coordinates and generated output videos

overlying skeletons on the input. Four frameworks saved only output videos, and another four saved neither keypoint coordinates nor output videos (see appendix Figure 8).

D. Models and Versions

For each framework, the study used the most current version available at the time of the experiments and employed the model provided by the developers. When multiple models were available, the selection followed the developers' recommendations, choosing the model expected to deliver the highest performance. If a framework offered both 2D and 3D models, we used the best-performing 3D model. Wherever possible, existing demo code ran the experiments. The experiments took place in February 2024 (Detectron2, MediaPipe, MHFormer, OpenPifPaf-vita, PoseFormerV2, StridedTransformer-Pose3D, YOLOv7), May 2024 (MeTR bs, MMPose, OpenPose, ultralytics), July 2024 (AlphaPose, OpenPifPaf), and August 2024 (MoveNet, rtmLib, ViTPose) (see Figure 1 for models and appendix Figure 7 for details on hardware and software environments).

Frameworks	Version	Used Model
AlphaPose	https://github.com/MVIG-SJTU/AlphaPose/tree/c60106d	multi_domain_fast50_regression_256x192.pth
Detectron2	0.6	keypoint_rcnn_X_101_32x8d_FPN_3x.yaml
MediaPipe	0.10.9	pose_landmarker_heavy.task
MeTRAbs	https://github.com/isarandi/metrabs/tree/8b2b116	metrabs_eff2l_y4
MHFormer	https://github.com/Vegetebird/MHFormer/tree/t45cc7b	351/model_4294.pth
MMPose	1.3.1	motionbert_dstformer-ft-243frm_8xb32-120e_h36m
MoveNet	https://github.com/tensorflow/docs/tree/15f45f8	movenet_thunder
OpenPifPaf	0.13.11	shufflenetv2k30
OpenPifPaf-vita	0.14.1	shufflenetv2k30
OpenPose	1.7.0	--net_resolution "1312x736" --scale_gap 0.25
PoseFormerV2	https://github.com/KitaoZhao/PoseFormerV2/tree/c2573ec	27_243_45.2.bin
rtmLib	0.0.12	mode='performance'
StridedTransformer-Pose3D	https://github.com/Vegetebird/StridedTransformer-Pose3D/tree/d30838	no_refine_4365.pth
ultralytics (YOLOv8)	8.2.22	YOLOv8x-pose-p6.pt
ViTPose	https://github.com/RizwanMunawar/ViTPose/tree/d521645	vitpose-h.pth
YOLOv7	https://github.com/RizwanMunawar/yolov7-pose-estimation/tree/60e65b2	yolov7-w6-pose.pt

Fig. 1. Frameworks, versions, and models used in the experiments. The study employed the most up-to-date framework versions available at the time of evaluation, using the developer-provided models. When the developers offered several models, we selected the one they recommended for the highest expected performance. For frameworks offering both 2D and 3D variants, we selected the best-performing 3D model.

E. Dataset

Our experiments used the dataset introduced by Reimer et al. (2022) and refer to their paper for details on dataset generation and description. We did not extend the dataset. It contains high-quality data recorded with Vicon, a gold-standard motion tracking system. This setup allowed precise calculation and comparison of framework performance. The dataset has been unpublished, ensuring a fair comparison because none of the frameworks had access to it during training. Originally, the dataset aimed to compare RKit, Apple's mobile 3D HPE framework [58], with Vicon in a controlled laboratory setting. During recording, Vicon captured full-body motion, while two iPads simultaneously recorded videos from distinct angles. The iPads recorded at 30 Hz with a resolution of 1920×1080, while the Vicon system captured data at 250 Hz [37].

Their dataset includes 10 participants (six males and four females) aged 22 – 31 years (mean 25.7 years). Heights ranged from 156 cm to 198 cm (mean 176 cm), weights from 53 kg to 90 kg (mean 69.5 kg), and body mass indices fell within 20.4 – 25.5. All participants had light skin coloration, were physically healthy without orthopedic or neurological impairments, and wore short, tight clothing (see Figure 13 in Reimer et al. (2022) for an example image from the dataset) [37].

The dataset features eight exercises targeting various body parts: front lunge, jumping jacks, lateral arm raise, leg extension crunch, reverse fly, side squat, single-leg deadlift, and squat. The participants repeated each exercise ten times in a video. They recorded videos from two views, frontal and 30° to the side (see Figure 2 in Reimer et al. (2022) [37]), producing 80 CSV files (marker positions for ten participants across eight exercises), 80 XCP files (Vicon metadata), and 152 videos (ten participants, eight exercises, two camera views; for one participant, only one camera view was available).

Due to data protection, we were allowed to include the data of nine (five male & four female) of the ten participants for this study.

The original dataset included video and Vicon data beyond the actual exercise durations. For this study, we manually trimmed all recordings to include only the exercises themselves, simplifying synchronization with Vicon data, since the video files lacked timestamps. Exercise durations ranged from 11 to 53 seconds.

F. Data Integration and Preparation

To compare results across frameworks, the analysis followed three main steps:

- 1) **Data Combination:** We merged all framework results and Vicon data into a single data frame for analysis per framework, arranging time in rows and joint data (including metadata) in columns. The two recording systems sampled at different rates (iPads 30 Hz, Vicon 250 Hz). To align the datasets, we used the nearest relative timestamp as the primary criterion for merging Vicon data with framework results.
- 2) **Data Synchronization:** Since the datasets lacked synchronized timestamps, we synchronized the recordings using landmark angles (a specific joint angle) for each exercise and camera view. We computed these angles for every result from every framework (see Section IV-G). We excluded Framework or Vicon results with fewer than 100 frames, as short recordings did not allow reliable synchronization. We calculated a shift factor for each video based on the maximum Pearson correlation between the landmark angle time series from each framework and those from Vicon. We limited the shift to ± 60 frames, assuming optimal alignment occurred when linear trends matched closely (see Figure 3 in Reimer et al. (2022) [37]). To reduce erroneous shift values from individual frameworks, we determined the final shift for each video as the median of the 16 shift values provided by all frameworks. We then applied this shift to synchronize the video data.

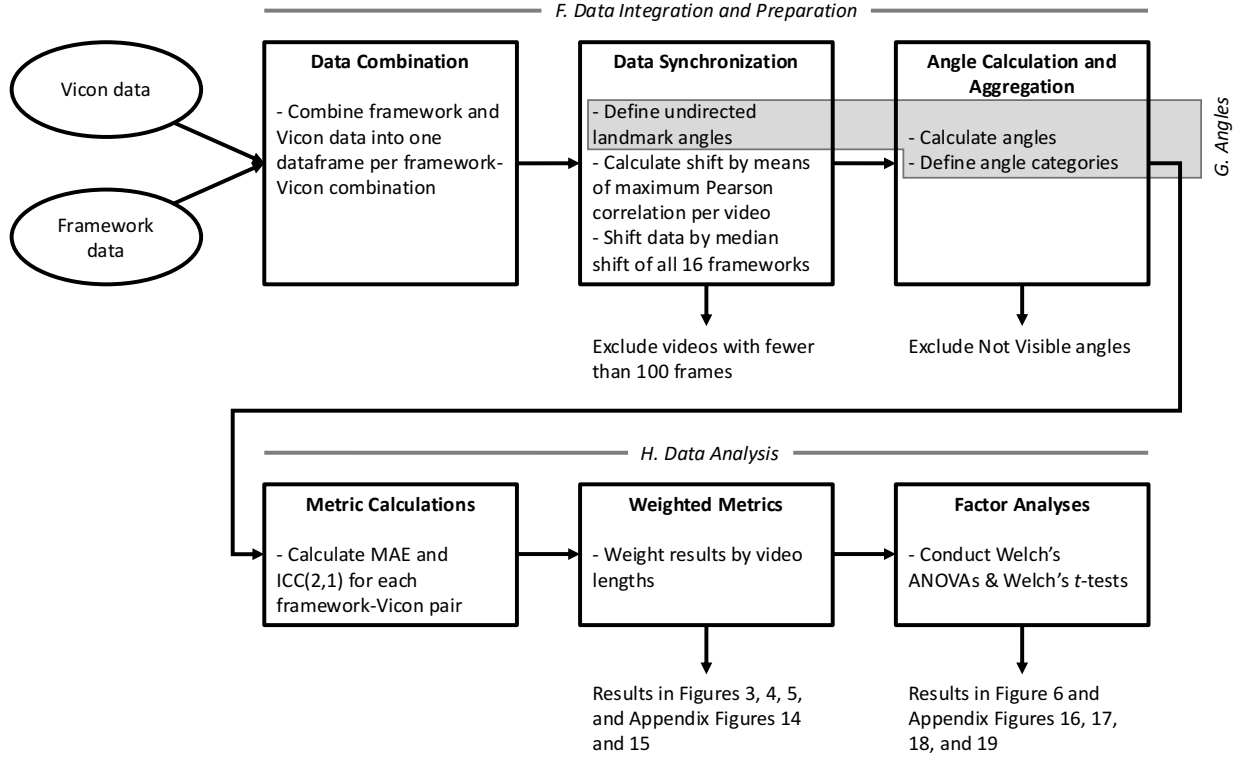


Fig. 2. Processing pipeline summarizing the steps described in Sections IV-F, IV-G, and IV-H

3) **Angle Calculation and Aggregation:** For the synchronized videos, we calculated the angles of interest following the procedure in Section IV-G. We aggregated angles into five angle categories: Side View, Frontal View, Oblique View, Not Visible, and No motion, based on visibility and camera perspective (see Section IV-G and Appendix Figure 12). We excluded the Not Visible category from the analysis to prevent distortions in the aggregated results. This process enabled the calculation of weighted Means, standard deviations, and weighted ICC(2,1) values for each perspective.

G. Angles

We calculated all joint angles as undirected angles in the interval $[0^\circ, 180^\circ]$ (see Formula 1) [37]. We defined the two vectors by the 3D coordinates of three joints A , B , and C (see Formulas 2 and 3) [37]. This definition places 2D frameworks at a systematic disadvantage in scenarios where depth information contributes substantially to the true joint angle, because 2D projections cannot capture out-of-plane motion present in the 3D reference data.

$$= \arccos \frac{\vec{v}_1 \cdot \vec{v}_2}{\|\vec{v}_1\|_2 \|\vec{v}_2\|_2} \quad (1)$$

$$\vec{v}_1 = A - B \quad (2)$$

$$\vec{v}_2 = C - B \quad (3)$$

We defined landmark angles to calculate the mutual Pearson correlation coefficients between every framework and the Vicon data for the Data Synchronization step (see Section

IV-F). For every exercise, we selected an angle with significant positive and negative radial acceleration (see Appendix Figure 9). This choice ensured that the maximal correlation coefficient corresponded to the correct shift value. We assumed that the framework and Vicon time series aligned best when their linear trends exhibited similar behavior.

During the reverse fly exercise, landmark angles sometimes “flipped around,” decreasing again within a single iteration (see Appendix Figure 10). This effect arose because the undirected angle calculation always selected the smallest angle between two vectors. When a limb crossed the body’s plane, the angle switched to the opposite side, creating a continuous decrease instead of a continuous increase. Since the calculation was in 3D, flipping occurred at different angles depending on orientation and could happen at angles well below 180° . This behavior did not affect data synchronization because we only required the optimal shift values, not the absolute correlation values. However, it posed a problem for subsequent calculations. We therefore excluded this angle definition from the final analysis. Instead, we calculated results using the definitions from Reimer et al. (2022) (see Appendix Figure 11), which remained stable throughout the exercises.

To make results comparable, we grouped the exercise and camera view combinations into five categories that reflect different perspectives on each joint (see Appendix Figure 12). For 2D frameworks, these perspectives are critical because they cannot account for depth.

Two reviewers independently assigned each angle in each exercise and camera view combination to one of the five angle categories. They resolved disagreements through discussion.

They assigned Side View when the vectors were primarily visible from the side with similar depth coordinates. They assigned Frontal View when the vectors were visible mainly from the front and ran essentially in depth-coplanarity. They assigned Oblique View to intermediate cases between side and front perspectives. They labeled a joint as No c tion when its angle remained nearly constant across repetitions. They labeled a joint as Not Visible when one of the two body parts forming the vectors adjacent to the angle was generally not visible.

These categories condensed the 16 combinations of eight exercises and two camera views into five approximately consistent groups. It is important to note that not all categories applied to every joint angle (see ppendix Figure 13). We excluded Not Visible from the analysis, while ll Views refers to all categories except Not Visible.

H. Data nalysis

We analyzed the data in three main steps:

- 1) **Metric Calculations:** For each pair of framework time series $\hat{y}$ and Vicon series y with n frames, we computed two metrics: M E and ICC(2,1) [50]. M E quantified absolute errors (using the abs and mean functions of NumPy [59])

$$M\ E = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|. \quad (4)$$

ICC(2,1) assessed correlation between the systems (using the intraclass_corr function of Pingouin [60], [61])

$$ICC(2,1) = \frac{MS_B - MS_E}{MS_B + (k - 1)MS_E + k \frac{MS_J - MS}{n}}. \quad (5)$$

where MS_B is the mean square between targets, MS_J the mean square between raters, MS_E the mean square residual, k the number of raters, and n the number of targets.

Each framework could provide up to (136 videos - 2 too short Vicon landmark angle) · 8 angles - 2 missing coordinates for a joint from Vicon - 60 Not Visible joints (up to seven per subject) = 1,010. OpenPose provided 1,002 measurements; all others provided the full 1,010.

- 2) **Weighted Metrics:** We grouped angle-video pairs into Side View, Frontal View, Oblique View, Not Visible, and No c tion (ppendix Figure 12). Let $\mathcal{A}_c$ be the set of all angle-pairs of time series combinations that are in $C = \{ll, Side\ View\}$. Let $m(a, v)$ be the metric of angle a in a pair of time series v . The weighted metric x_c is calculated by

$$x_c = \frac{\sum_{(a, v) \in \mathcal{A}_c} m(a, v) \cdot w_v}{\sum_{(a, v) \in \mathcal{A}_c} w_v}. \quad (6)$$

- 3) **Factor nalyses:** We applied Welch's NOV s [62] and Welch's t -tests [63] to logarithmically transformed M E values. We selected them because preliminary data exploration indicated unequal sample sizes and unequal

variances across groups, making traditional NOV [64] and Student's t -test [63] less reliable. Logarithmic transformation reduced skewness and dampened outliers. Both tests assume independence and approximate normality within groups. The data did not guarantee full independence due to simultaneous recordings across camera views and joints, but correlations should be small given partial independence and aggregation.

Welch's NOV tested the effect of angle, exercise, subject, and angle category, where each categorical variable had more than two groups. We selected the first three categorical variables as potential influencing factors to ensure comparability with the paper by Reimer et al. (2022) [37]. We selected angle category to determine the possible influence of the kind of view on the joint. We computed Welch's NOV by means of Pingouin's welch_anova function³ [60], [65]. For group i , with variance s_i^2 and sample size n_i , the weight was

$$w_i = \frac{n_i}{s_i^2}. \quad (7)$$

The weighted mean was

$$\bar{Y}_{\text{welch}} = \frac{\sum_{i=1}^r w_i \bar{Y}_i}{\sum_{i=1}^r w_i}, \quad (8)$$

where $\bar{Y}_i$ is the sample mean and r the number of categories. The adjusted between-group sum of squares was

$$SS_{\text{between, adj}} = \sum_{i=1}^r w_i (\bar{Y}_i - \bar{Y}_{\text{welch}})^2. \quad (9)$$

Welch's correction factor was

$$\Lambda = \frac{3 \sum_{i=1}^r \left(\frac{1}{n_i} - \frac{1}{n} \right)}{r^2 - 1} \frac{1}{\sum_{i=1}^r w_i}, \quad (10)$$

given the F-statistic

$$F_{\text{welch}} = \frac{SS_{\text{between, adj}} / (r - 1)}{1 + \frac{2}{3} \frac{(r - 2)}{r}}, \quad (11)$$

with degrees of freedom

$$df_{\text{denominator}} = \frac{1}{\Lambda} \quad \text{and} \quad df_{\text{numerator}} = r - 1. \quad (12)$$

Welch's t -test compared logarithmically transformed M Es across camera view, center moved, and sex. For groups with means $\bar{x}_1, \bar{x}_2$, variances s_1^2, s_2^2 , and sizes n_1, n_2 , we computed

Welch's t -test compared logarithmically transformed M Es across camera view, center moved, and sex. We selected the first two categorical variables as potential influencing factors to facilitate comparison with the paper by Reimer et al. (2022) [37]. We selected sex to determine possible gender biases within the frameworks. We computed Welch's t -tests by means of Pingouin's

³Pingouin's welch_anova function, available at: https://pingouin-stats.org/build/html/generated/pingouin.welch_anova.html

ttest function with correction, which is an implementation of it ⁴ [60], [66]. For groups with sample means $\bar{x}_1, \bar{x}_2$, variances s_1^2, s_2^2 , and sizes n_1, n_2 , the T-statistic was:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}, \quad (13)$$

with degrees of freedom

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}}. \quad (14)$$

V. RESULTS

A. All Views Framework Performance

To answer RQ 1—How accurate are the chosen frameworks compared to Vicon?—We analyzed weighted M E and weighted ICC(2,1) across all exercises and camera views.

Overall, the 3D frameworks outperformed the 2D frameworks. MeTR bs demonstrated the highest accuracy and reliability, showing consistently low errors and strong agreement with the reference data. Other 3D models, such as MHFormer and MMPose, performed moderately well, while MediaPipe exhibited substantially higher errors and lower reliability. The 2D frameworks, in contrast, showed larger errors and weaker agreement with Vicon. Among them, rtllib produced the best overall results, whereas ultralytics performed the poorest.

Aggregated performance across all views confirmed that 3D frameworks generally provide a clear advantage over 2D approaches (see Figure 3).

	Frameworks	All Views weighted MAE	All Views weighted ICC(2,1)	Side View weighted MAE	Side View weighted ICC(2,1)
2D Output	AlphaPose	21.48 ± 19.26	0.38 ± 0.39	7.3 ± 3.94	0.89 ± 0.2
	Detectron2	22.05 ± 18.44	0.36 ± 0.39	8.32 ± 4.77	0.87 ± 0.23
	MoveNet	29.41 ± 21.16	0.27 ± 0.34	17.69 ± 6.02	0.74 ± 0.25
	OpenPifPaf	23.38 ± 19.14	0.35 ± 0.38	8.88 ± 5.77	0.86 ± 0.23
	OpenPifPaf-vita	23.39 ± 19.14	0.35 ± 0.38	8.88 ± 5.76	0.86 ± 0.23
	OpenPose	24.05 ± 19.0	0.34 ± 0.37	11.14 ± 6.06	0.82 ± 0.23
	rtllib	21.33 ± 18.97	0.39 ± 0.39	7.62 ± 4.26	0.89 ± 0.19
	ultralytics (YOLOv8)	29.97 ± 20.89	0.25 ± 0.33	19.05 ± 7.78	0.7 ± 0.25
	YOLOv7	22.17 ± 19.2	0.36 ± 0.38	8.13 ± 4.35	0.89 ± 0.18
3D Output	MediaPipe	17.8 ± 9.64	0.38 ± 0.4	17.63 ± 7.99	0.69 ± 0.3
	MeTRAbs	7.06 ± 4.16	0.58 ± 0.4	4.22 ± 2.15	0.95 ± 0.12
	MHFormer	12.36 ± 9.22	0.47 ± 0.42	7.09 ± 3.84	0.89 ± 0.22
	MMPose	13.92 ± 11.47	0.46 ± 0.43	7.06 ± 4.37	0.91 ± 0.19
	PoseFormerV2	13.38 ± 11.37	0.44 ± 0.42	8.53 ± 5.31	0.87 ± 0.21
	StridedTransformer-Pose3D	16.55 ± 14.73	0.38 ± 0.41	11.63 ± 5.87	0.79 ± 0.27
	VITPose	16.74 ± 14.76	0.37 ± 0.4	10.52 ± 5.74	0.82 ± 0.22

Fig. 3. Weighted M E with standard deviation in degrees and weighted ICC(2,1) with standard deviation per framework for All Views and for Side View only

B. Side View Framework Performance

To address RQ 2—Can 2D frameworks perform comparably to 3D frameworks in Side View scenarios where depth

information is significantly limited?—We also focused on the Side View specifically.

Because the Side View reduces depth-related ambiguities, it provides a favorable condition for 2D pose estimation. When evaluated in this setting, both 2D and 3D frameworks showed improved performance. Although 3D frameworks again achieved the highest accuracy, several 2D methods approached 3D performance under these conditions. In particular, lphaPose, rtllib, and YOLOv7 displayed substantial improvements in both error and consistency. Nevertheless, the best 3D framework, MeTR bs, maintained a noticeable performance advantage. Conversely, ultralytics continued to show comparatively weak results, even in Side View. Taken together, these findings suggest that although Side View reduces the performance gap between 2D and 3D methods, 3D frameworks remain more accurate overall (see Figure 3).

C. Framework Performance across Joints and Views

To provide a more comprehensive answer to RQ 1 and RQ 2, we analyzed the results for each angle with respect to M E and ICC(2,1).

Across all evaluated joints and camera views, the relative performance rankings of the frameworks remained comparable with those observed in the aggregated analyses. Among the 3D frameworks, MeTR bs achieved the lowest errors and the highest reliability across nearly every joint, regardless of viewing condition. MHFormer and MMPose followed as the next-best performers, although their results varied more across joints. In contrast, MediaPipe showed higher errors and lower agreement with Vicon for most joints, particularly in views involving occlusions or complex limb orientations (see Figures 4 and 5 as well as Appendix Figures 14 and 15).

Within the 2D category, performance varied across joints and views. lphaPose, rtllib, and YOLOv7 demonstrated the strongest results, especially in Side View, where their results approached those of the better-performing 3D frameworks (see Figures 4 and 5). However, all 2D frameworks showed lower performance in Frontal and Oblique Views (see Appendix Figures 14 and 15).

D. Factor Analysis

To answer RQ 3—What are the main factors influencing HPE performance?—We conducted Welch's NOV s for multi-group factors (exercise, angle, subject, and angle category) and Welch's *t*-tests for binary factors (camera view, center moved, and sex) on the weighted logarithmic M Es.

Across all views, the exercise and angle factors had statistically significant effects ($p < 0.01$), indicating that some movements and joints increase estimation difficulty regardless of framework. The subject factor showed no significant influence ($p \geq 0.05$) for any framework except VITPose, suggesting good generalizability across participants. The angle category factor had statistically significant effects ($p < 0.01$) for all frameworks except MediaPipe ($p \geq 0.05$). However, the corresponding effect sizes (η^2) were generally smaller for the 3D frameworks (see Appendix Figure 16).

⁴Pingouin's ttest function, available at: <https://pingouin-stats.org/build/html/generated/pingouin.ttest.html>

	Framework	Angle Category	leftElbow	rightElbow	leftShoulder	rightShoulder	leftHip	rightHip	leftKnee	rightKnee
2D Output	AlphaPose	All Views	25.39 ± 19.33	27.14 ± 22.08	26.04 ± 21.76	20.78 ± 19.76	21.23 ± 18.73	20.87 ± 20.58	15.61 ± 13.58	16.71 ± 13.9
		Side View			6.32 ± 1.56	5.41 ± 2.5	9.76 ± 3.61	6.07 ± 2.95	6.04 ± 2.76	9.24 ± 5.56
	Detectron2	All Views	26.14 ± 17.98	27.29 ± 19.58	25.13 ± 21.19	20.6 ± 19.17	22.77 ± 18.69	21.35 ± 19.59	16.54 ± 14.16	18.2 ± 13.96
		Side View			6.77 ± 1.44	5.59 ± 2.44	12.05 ± 5.15	6.56 ± 2.53	7.27 ± 3.09	10.06 ± 6.47
	MoveNet	All Views	27.69 ± 20.85	28.95 ± 22.7	38.74 ± 26.18	35.45 ± 23.08	29.17 ± 20.38	31.26 ± 20.05	21.5 ± 15.45	24.77 ± 15.19
		Side View			15.68 ± 3.07	16.31 ± 2.17	16.2 ± 5.21	21.06 ± 4.37	13.69 ± 4.93	24.21 ± 6.63
	OpenPifPaf	All Views	33.34 ± 21.5	29.84 ± 21.15	25.12 ± 19.91	19.71 ± 18.19	23.74 ± 17.74	21.89 ± 19.8	17.76 ± 14.73	17.98 ± 13.95
		Side View			6.73 ± 1.5	5.45 ± 2.53	13.59 ± 7.12	6.65 ± 2.88	8.05 ± 4.36	10.61 ± 6.6
	OpenPifPaf-vita	All Views	33.36 ± 21.51	29.83 ± 21.15	25.13 ± 19.91	19.7 ± 18.18	23.74 ± 17.74	21.89 ± 19.8	17.76 ± 14.73	17.98 ± 13.94
		Side View			6.73 ± 1.5	5.45 ± 2.53	13.58 ± 7.11	6.65 ± 2.88	8.05 ± 4.36	10.61 ± 6.6
	OpenPose	All Views	33.34 ± 21.93	29.96 ± 21.64	27.28 ± 19.55	20.96 ± 17.79	23.32 ± 17.62	21.99 ± 19.46	19.2 ± 14.72	18.95 ± 13.54
		Side View			10.46 ± 4.72	8.51 ± 4.91	14.66 ± 7.08	9.18 ± 4.12	10.72 ± 6.09	11.96 ± 5.51
3D Output	rtmlib	All Views	24.44 ± 19.53	26.46 ± 19.46	25.09 ± 22.21	20.85 ± 20.95	21.38 ± 18.72	20.94 ± 19.74	15.74 ± 14.05	17.35 ± 14.06
		Side View			6.34 ± 1.58	5.71 ± 2.42	9.39 ± 3.79	7.02 ± 3.1	6.11 ± 3.3	10.36 ± 6.26
	ultralitics (YOLOv8)	All Views	28.66 ± 21.42	28.03 ± 24.38	36.92 ± 24.12	33.59 ± 19.66	30.66 ± 21.8	33.33 ± 20.86	21.56 ± 16.16	28.7 ± 14.27
		Side View			15.61 ± 2.41	15.64 ± 2.94	16.71 ± 4.81	24.78 ± 5.23	13.16 ± 5.98	29.96 ± 5.57
	YOLOv7	All Views	24.24 ± 20.23	27.42 ± 22.49	26.53 ± 21.48	21.16 ± 19.01	22.76 ± 18.75	22.41 ± 19.73	16.26 ± 14.23	18.1 ± 14.26
		Side View			7.5 ± 2.87	6.5 ± 2.93	10.07 ± 4.62	7.26 ± 2.78	5.79 ± 2.28	11.18 ± 5.68
	MediaPipe	All Views	21.6 ± 11.14	20.5 ± 11.16	14.2 ± 6.39	16.08 ± 9.06	14.27 ± 6.03	15.52 ± 7.02	21.06 ± 10.79	18.84 ± 10.05
		Side View			11.58 ± 3.89	12.21 ± 5.23	18.42 ± 4.83	23.06 ± 6.15	18.7 ± 7.61	20.71 ± 10.81
	MeTRbs	All Views	9.29 ± 4.63	10.45 ± 4.3	7.38 ± 4.0	7.06 ± 3.91	6.1 ± 3.02	5.9 ± 3.15	5.48 ± 3.33	5.31 ± 3.74
		Side View			5.28 ± 1.45	4.76 ± 1.95	4.27 ± 2.19	4.98 ± 3.21	3.56 ± 1.48	3.31 ± 1.73
	MHFormer	All Views	15.67 ± 8.34	14.52 ± 8.55	10.71 ± 10.01	10.8 ± 9.31	9.33 ± 5.84	9.76 ± 6.72	14.52 ± 10.9	13.69 ± 10.63
		Side View			5.03 ± 1.77	5.02 ± 1.52	7.03 ± 2.99	6.57 ± 3.71	10.18 ± 4.56	7.23 ± 4.07
	MMPose	All Views	23.15 ± 15.85	20.33 ± 13.45	14.57 ± 12.45	12.6 ± 11.15	10.27 ± 6.81	10.34 ± 8.54	11.22 ± 6.92	10.75 ± 6.77
		Side View			5.96 ± 2.53	6.12 ± 4.44	9.17 ± 3.78	4.18 ± 2.57	7.0 ± 4.4	8.2 ± 5.02
3D Output	PoseFormerV2	All Views	13.37 ± 6.69	14.4 ± 11.91	12.37 ± 11.56	11.55 ± 10.34	11.53 ± 10.33	12.42 ± 11.73	15.58 ± 11.83	15.51 ± 13.77
		Side View			6.15 ± 3.64	5.48 ± 2.72	7.21 ± 3.07	7.06 ± 3.89	13.74 ± 7.1	9.49 ± 3.79
	StridedTransformer-Pose3D	All Views	18.35 ± 12.3	17.22 ± 12.69	17.86 ± 19.42	16.23 ± 17.74	14.56 ± 12.58	14.47 ± 14.49	17.74 ± 13.1	16.67 ± 14.46
		Side View			10.33 ± 5.0	10.14 ± 4.14	11.19 ± 5.34	9.42 ± 4.58	14.13 ± 6.77	13.05 ± 6.64
	ViTPose	All Views	17.58 ± 12.82	17.2 ± 10.66	20.71 ± 18.99	16.9 ± 18.62	14.18 ± 12.43	15.54 ± 14.24	16.35 ± 13.37	16.73 ± 15.21
		Side View			8.88 ± 5.48	7.15 ± 3.61	11.57 ± 7.37	9.56 ± 5.08	12.9 ± 6.27	11.57 ± 2.4
	Framework	Angle Category	leftElbow	rightElbow	leftShoulder	rightShoulder	leftHip	rightHip	leftKnee	rightKnee

Fig. 4. Weighted M Es (in °) and standard deviations for eight joint angles across all frameworks, shown for All Views and Side View. Colors indicate relative error magnitude from low (green) to high (red). Results for leftElbow and rightElbow are not shown in Side View because no videos contained these joints in Side View (see Appendix Figure 13).

When only the results of the angle category Side View were considered, the angle and exercise factors remained statistically significant for nearly all frameworks ($p < 0.05$). OpenPose and StridedTransformer-Pose3D were exceptions, for which the angle factor did not reach significance ($p \geq 0.05$). The subject factor again showed no significant influence ($p \geq 0.05$) in Side View for all frameworks except ViTPose (see Appendix Figure 17).

For the binary factors across all views, the camera view factor (frontal vs. side) had statistically significant effects ($p < 0.01$) for all 3D frameworks except MMPose and for several 2D frameworks ($p < 0.05$), confirming that viewpoint strongly affects performance. The center moved factor (movement vs. no movement) showed statistically significant effects ($p < 0.02$) for all frameworks except MeTRbs ($p \geq 0.05$), indicating that spatial drift tends to degrade performance (see Appendix Figure 18). The sex factor (female vs. male) was statistically significant ($p < 0.05$) only for MoveNet, ultralytics, and ViTPose (see Figure 6).

When only considering Side View, the camera view factor remained statistically significant ($p < 0.05$) for most frameworks (see Appendix Figure 19). The center moved factor showed statistically significant differences ($p < 0.05$) only for YOLOv7, MMPose, StridedTransformer-Pose3D, and ViTPose. The sex factor was statistically significant ($p < 0.05$) for MeTRbs and ViTPose.

VI. DISCUSSION

No single framework excelled across all joint angles and perspectives. Instead, performance varied across joints and views. However, the results enable practitioners to adjust exercises according to their chosen framework or select the most suitable framework based on the joint angles of interest and perspective (see Appendix Figures 14 and 15).

Overall, MeTRbs emerged as the most reliable framework, showing the lowest M Es and the highest ICC(2,1) across multiple angles and angle categories. Notably, MeTRbs and the strongest 2D frameworks—AlphaPose, rtmlib, and YOLOv7—used top-down approaches, while rtmlib and MMPose, the next-best 3D frameworks, relied on two-stage approaches. This pattern suggests that top-down architectures remain the most dependable choice when robust performance is required.

An interesting and practically relevant finding emerged in the Side View analyses: 2D frameworks performed on par with 3D models. This indicates that in settings where depth displacement is minimal—such as rehabilitation, physiotherapy, gymnastics, or fitness exercises in controlled sagittal-plane conditions—2D frameworks like AlphaPose, rtmlib, or YOLOv7 may provide sufficiently accurate solutions. In contrast, 2D frameworks underperformed in Oblique and Frontal Views because they were unable to estimate depth. Conse-

Framework		Angle Category	leftElbow	rightElbow	leftShoulder	rightShoulder	leftHip	rightHip	leftKnee	rightKnee
2D Output	AlphaPose	All Views	0.05 ± 0.14	0.09 ± 0.19	0.46 ± 0.42	0.51 ± 0.38	0.47 ± 0.39	0.5 ± 0.38	0.44 ± 0.38	0.52 ± 0.39
		Side View			0.98 ± 0.01	0.97 ± 0.04	0.79 ± 0.3	0.96 ± 0.03	0.81 ± 0.23	0.95 ± 0.08
	Detectron2	All Views	0.05 ± 0.14	0.09 ± 0.2	0.46 ± 0.43	0.52 ± 0.38	0.41 ± 0.39	0.48 ± 0.38	0.4 ± 0.39	0.46 ± 0.4
		Side View			0.98 ± 0.01	0.97 ± 0.03	0.78 ± 0.24	0.95 ± 0.03	0.72 ± 0.36	0.93 ± 0.09
	MoveNet	All Views	0.02 ± 0.13	0.03 ± 0.12	0.38 ± 0.41	0.37 ± 0.38	0.34 ± 0.34	0.32 ± 0.29	0.34 ± 0.35	0.38 ± 0.32
		Side View			0.92 ± 0.04	0.83 ± 0.19	0.68 ± 0.29	0.73 ± 0.08	0.64 ± 0.34	0.74 ± 0.16
	OpenPifPaf	All Views	0.04 ± 0.12	0.09 ± 0.2	0.44 ± 0.42	0.54 ± 0.37	0.38 ± 0.35	0.45 ± 0.39	0.38 ± 0.37	0.47 ± 0.38
		Side View			0.98 ± 0.01	0.97 ± 0.03	0.71 ± 0.29	0.95 ± 0.03	0.73 ± 0.3	0.92 ± 0.1
	OpenPifPaf-vita	All Views	0.04 ± 0.12	0.09 ± 0.2	0.44 ± 0.42	0.54 ± 0.37	0.38 ± 0.35	0.45 ± 0.39	0.38 ± 0.37	0.47 ± 0.38
		Side View			0.98 ± 0.01	0.97 ± 0.03	0.71 ± 0.29	0.95 ± 0.03	0.73 ± 0.3	0.92 ± 0.1
	OpenPose	All Views	0.04 ± 0.13	0.09 ± 0.19	0.43 ± 0.41	0.51 ± 0.37	0.42 ± 0.34	0.44 ± 0.37	0.35 ± 0.35	0.44 ± 0.38
		Side View			0.96 ± 0.02	0.94 ± 0.06	0.71 ± 0.28	0.88 ± 0.18	0.67 ± 0.27	0.89 ± 0.16
	rtmlib	All Views	0.06 ± 0.14	0.1 ± 0.21	0.47 ± 0.43	0.54 ± 0.38	0.45 ± 0.39	0.48 ± 0.39	0.44 ± 0.39	0.52 ± 0.38
		Side View			0.98 ± 0.01	0.97 ± 0.04	0.81 ± 0.27	0.95 ± 0.04	0.79 ± 0.26	0.93 ± 0.08
	ultralitics (YOLOv8)	All Views	0.02 ± 0.12	0.02 ± 0.12	0.39 ± 0.41	0.36 ± 0.39	0.32 ± 0.36	0.28 ± 0.27	0.32 ± 0.35	0.29 ± 0.28
		Side View			0.92 ± 0.03	0.83 ± 0.22	0.68 ± 0.27	0.61 ± 0.15	0.64 ± 0.33	0.61 ± 0.14
	YOLOv7	All Views	0.07 ± 0.14	0.08 ± 0.18	0.47 ± 0.42	0.51 ± 0.38	0.41 ± 0.38	0.44 ± 0.38	0.42 ± 0.38	0.48 ± 0.38
		Side View			0.98 ± 0.01	0.95 ± 0.07	0.79 ± 0.27	0.94 ± 0.05	0.82 ± 0.21	0.92 ± 0.08
Framework		Angle Category	leftElbow	rightElbow	leftShoulder	rightShoulder	leftHip	rightHip	leftKnee	rightKnee
3D Output	MediaPipe	All Views	0.03 ± 0.16	0.03 ± 0.14	0.58 ± 0.37	0.46 ± 0.39	0.5 ± 0.43	0.5 ± 0.38	0.43 ± 0.38	0.49 ± 0.39
		Side View			0.95 ± 0.03	0.85 ± 0.18	0.54 ± 0.42	0.65 ± 0.14	0.54 ± 0.27	0.76 ± 0.19
	MeTRAbs	All Views	0.21 ± 0.25	0.18 ± 0.25	0.77 ± 0.3	0.76 ± 0.26	0.71 ± 0.36	0.71 ± 0.36	0.65 ± 0.43	0.67 ± 0.41
		Side View			0.99 ± 0.01	0.96 ± 0.07	0.87 ± 0.23	0.97 ± 0.03	0.95 ± 0.07	0.99 ± 0.01
	MHFormer	All Views	0.1 ± 0.2	0.2 ± 0.27	0.61 ± 0.44	0.62 ± 0.38	0.62 ± 0.38	0.63 ± 0.38	0.44 ± 0.44	0.5 ± 0.42
		Side View			0.99 ± 0.01	0.95 ± 0.11	0.8 ± 0.31	0.95 ± 0.04	0.78 ± 0.29	0.96 ± 0.04
	MMPose	All Views	0.07 ± 0.18	0.12 ± 0.24	0.55 ± 0.44	0.57 ± 0.4	0.56 ± 0.43	0.62 ± 0.39	0.52 ± 0.44	0.61 ± 0.41
		Side View			0.98 ± 0.01	0.96 ± 0.05	0.75 ± 0.35	0.98 ± 0.02	0.89 ± 0.1	0.95 ± 0.05
	PoseFormerV2	All Views	0.11 ± 0.19	0.18 ± 0.25	0.6 ± 0.44	0.61 ± 0.37	0.55 ± 0.41	0.57 ± 0.41	0.41 ± 0.43	0.49 ± 0.42
		Side View			0.98 ± 0.02	0.96 ± 0.07	0.8 ± 0.32	0.95 ± 0.04	0.73 ± 0.25	0.93 ± 0.05
	StridedTransformer-Pose3D	All Views	0.13 ± 0.2	0.16 ± 0.25	0.56 ± 0.42	0.57 ± 0.38	0.41 ± 0.47	0.48 ± 0.43	0.34 ± 0.4	0.4 ± 0.41
		Side View			0.93 ± 0.06	0.86 ± 0.18	0.65 ± 0.43	0.88 ± 0.09	0.68 ± 0.25	0.85 ± 0.15
	ViTPose	All Views	0.09 ± 0.2	0.11 ± 0.25	0.46 ± 0.44	0.57 ± 0.38	0.43 ± 0.42	0.46 ± 0.4	0.38 ± 0.39	0.42 ± 0.38
		Side View			0.94 ± 0.06	0.92 ± 0.11	0.7 ± 0.33	0.88 ± 0.1	0.7 ± 0.23	0.88 ± 0.06
Framework		Angle Category	leftElbow	rightElbow	leftShoulder	rightShoulder	leftHip	rightHip	leftKnee	rightKnee

Fig. 5. Weighted ICC(2,1) values and standard deviations for eight joint angles across all frameworks, shown for All Views and Side View. Colors indicate relative agreement levels from low (red) to high (green). Results for leftElbow and rightElbow are not shown in Side View because no videos contained these joints in Side View (see Appendix Figure 13).

Framework		T	dof	Alternative	p-value	CI95%	Cohen-d	BF10	Power
2D Output	AlphaPose	1.28	956.96	two-sided	0.2	[-0.04, 0.19]	0.08	0.158	0.25
	Detectron2	0.91	965.08	two-sided	0.36	[-0.06, 0.16]	0.06	0.106	0.15
	MoveNet	2.09	919.91	two-sided	0.04	[0.01, 0.2]	0.13	0.607	0.56
	OpenPifPaf	1.84	967.04	two-sided	0.07	[-0.01, 0.22]	0.12	0.371	0.45
	OpenPifPaf-vita	1.84	967.15	two-sided	0.07	[-0.01, 0.22]	0.12	0.371	0.45
	OpenPose	1.54	974.28	two-sided	0.12	[-0.02, 0.18]	0.1	0.226	0.34
	rtmlib	1.23	973.32	two-sided	0.22	[-0.04, 0.18]	0.08	0.149	0.23
	ultralitics (YOLOv8)	2.02	944	two-sided	0.04	[0.0, 0.19]	0.13	0.523	0.53
	YOLOv7	1.44	953.24	two-sided	0.15	[-0.03, 0.18]	0.09	0.197	0.31
Framework		T	dof	Alternative	p-Value	CI95%	Cohen-d	BF10	Power
3D Output	MediaPipe	-1.81	1000.7	two-sided	0.07	[-0.13, 0.01]	0.11	0.352	0.44
	MeTRAbs	1.88	951.07	two-sided	0.06	[-0.0, 0.15]	0.12	0.404	0.47
	MHFormer	-0.19	979.49	two-sided	0.85	[-0.09, 0.08]	0.01	0.072	0.05
	MMPose	-0.11	952.39	two-sided	0.91	[-0.09, 0.08]	0.01	0.071	0.05
	PoseFormerV2	-0.04	961.92	two-sided	0.97	[-0.09, 0.09]	0	0.071	0.05
	StridedTransformer-Pose3D	-1.83	960.41	two-sided	0.07	[-0.18, 0.01]	0.12	0.366	0.45
	ViTPose	-2.08	954.19	two-sided	0.04	[-0.2, -0.01]	0.13	0.594	0.55
Framework		T	dof	Alternative	p-Value	CI95%	Cohen-d	BF10	Power

Fig. 6. Welch's *t*-test results for sex

quently, 3D frameworks such as MeTRAbs, MHFormer, or MMPose remain the preferred choice for applications requiring multi-angle or frontal analyses, including sports science, clinical gait analysis, human-robot interaction, surveillance, or virtual try-ons.

Statistical analyses, including Welch's *NOV*s and Welch's *t*-tests, highlighted several determinants of framework performance. Angle, exercise type, and angle category strongly influenced results, while subject variability had a negligible impact. Camera view and center moved also affected out-

comes, but to varying degrees depending on the framework. Sex generally had no effect, except for MoveNet, ultralytics, and ViTPose. These results emphasize the importance of experimental design: clinicians, coaches, and researchers must carefully consider camera setups and exercise types to minimize confounding effects and maximize reliability. Notably, because framework performance was largely unaffected by participant sex, most frameworks can be deployed without significant concerns about gender-related bias.

lthough we did not explicitly benchmark runtimes, we observed substantial differences during the experiments. Processing one hour of video ranged from 2 to 26 hours across frameworks. Importantly, runtime did not correlate with accuracy: slower frameworks were not necessarily more effective. The differences in server configurations across frameworks, as outlined in the technical details (see Appendix Section VIII), should be considered when interpreting these results. Usability also varied. AlphaPose, OpenPifPaf, OpenPifPaf-vita, OpenPose, ultralytics, MHFormer, MMPose, and StridedTransformer-Pose3D provided fully functional demos that directly saved keypoint coordinates. The other frameworks required additional coding to enable this functionality. In practice, ease of implementation may be as decisive as performance when choosing a framework.

Comparison with related work (see Section III) proved difficult due to methodological differences. For performance

measurements, we could only compare our ranking with papers that evaluated at least two frameworks using similar angles. Our results align with the rankings presented by Wang et al. (2021), who ranked lphaPose above Detectron2 and OpenPose, aligning with our findings [54]. VanKeerbilck et al. (2022) ranked MediaPipe above OpenPose, consistent with our results across all angles [26]. Cariola et al. (2023) found MMPose best, followed by lphaPose, and MediaPipe [53], again consistent with our results despite using a different pre-trained model for MMPose. Murugan et al. (2024) also observed lower performance for MediaPipe compared to MMPose [51]. The only divergence came from Sabo et al. (2023), who reported no clear winner among lphaPose, Detectron2, MediaPipe, MoveNet, and OpenPose [55]. In our study, lphaPose performed best, followed by Detectron2, OpenPose, and MoveNet. Notably, none of the related works investigated sex effects. The only study to investigate any factors was conducted by Murugan et al. (2024), who identified a significant difference in MediaPipe performance across camera views [51], a finding also supported by our study.

In summary, our study identified actionable trends:

- For Side View analyses, 2D frameworks (lphaPose, rtmlib, YOLOv7) provide a practical alternative to 3D models.
- For multi-angle or frontal analyses, 3D frameworks (MeTR bs, MHFormer, MMPose) remain essential.
- For specific perspective angle combinations please see ppendix Figures 14 and 15.
- Top-down architectures delivered the strongest results across both 2D and 3D frameworks.
- Framework performance was largely unaffected by sex (except for MoveNet, ultralytics, and ViTPose).

Future research should expand evaluations to more diverse populations, to less controlled environments, and to include explicit runtime benchmarks. Testing frameworks in real-world applications will be necessary to fully validate and translate these findings into practice.

. Dataset Limitations

The study, however, inherits several limitations from the underlying dataset [37]. Data came from only nine participants under controlled laboratory conditions, limiting generalizability. Participant characteristics, such as a restricted height range, normal BMI, and uniform light skin tone, may introduce bias. Moreover, the exercises emphasized lower-body motion and frontal/right-side views, leaving some joints, particularly elbows and left-side joints, underrepresented.

In addition, the evaluation focuses on single-person exercise tasks with minimal occlusion. s a result, the findings may not fully extend to more dynamic or ecologically complex scenarios, such as multi-person interactions, environments with frequent self-occlusion, or naturalistic motion.

Nonetheless, the dataset and resulting analyses remain highly relevant for controlled motion settings, where movement is typically structured. Such environments include, but are not limited to, rehabilitation, physiotherapy, gymnastics,

fitness exercises, sports science, clinical gait analysis, human–robot interaction, and virtual try-on applications. In these domains, the controlled nature of the tasks aligns well with the conditions represented in our data, making the findings directly useful for practitioners and researchers working in these contexts.

B. Limitations of the Statistical Tests

The statistical analyses in this study relied on Welch's NOV and Welch's *t*-test applied to logarithmically transformed M E values. While we chose these tests for their robustness to unequal variances and sample sizes, several assumptions were only partially satisfied.

First, both Welch's NOV and Welch's *t*-test assume independence of observations. In our dataset, this assumption is only approximately met because joint angles were recorded simultaneously across camera views and exercises for the same participants. s a result, minor correlations between angles or perspectives cannot be excluded. lthough this dependency could, in principle, be modeled more precisely using mixed-effects approaches, the large number of partially independent observations and the aggregation across subjects and videos likely limited its practical impact on the reported results.

Second, these tests assume that the distributions within each group are approximately normal. Logarithmic transformation improved the symmetry of M E distributions and reduced the influence of extreme values, but it did not achieve perfect normality across all factors. However, Welch's tests remained reliable under moderate deviations from normality, particularly when sample sizes are large and unbalanced.

These limitations suggest that while the reported *p*-values should be interpreted cautiously, the main trends and relative differences among frameworks are expected to remain valid.

VII. CONCLUSION

This study provided a comprehensive evaluation of 2D and 3D HPE frameworks, using weighted M E and weighted ICC(2,1) as performance metrics. No framework excelled across the board, but MeTR bs consistently delivered the strongest results among 3D models, while lphaPose, rtmlib, and YOLOv7 led among 2D frameworks. Importantly, 2D frameworks matched 3D performance in Side View scenarios where depth information was less critical.

Statistical analyses revealed that framework performance depends primarily on angle, exercise type, angle category, camera view, and center moved. Subject variability and sex had little to no effect for most frameworks.

Despite limitations in dataset size, diversity, and laboratory constraints, the findings align with related studies and provide valuable insights into framework selection.

While this study focused on 16 frameworks (see Section IV-B), we remain open to testing additional frameworks upon reasonable request and to sharing extracted keypoint coordinates with the research community.

DECLARATION OF GENERATIVE AND ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, we used generative AI (Grammarly) to proofread the text and eliminate typos and grammatical flaws. After that, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

REFERENCES

- [1] L. Song, G. Yu, J. Yuan, and Z. Liu, "Human pose estimation and its application to action recognition: survey," *Journal of visual communication and image representation*, vol. 76, p. 103055, 2021.
- [2] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," *Advances in neural information processing systems*, vol. 25, 2012.
- [3] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *3rd International Conference on Learning Representations*. Computational and biological learning society, 2015.
- [4] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.
- [5] C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, and A. Rabinovich, "Going deeper with convolutions," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2015, pp. 1–9.
- [6] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "Imagenet: large-scale hierarchical image database," in *IEEE conference on computer vision and pattern recognition*. Ieee, 2009, pp. 248–255.
- [7] C. Ionescu, D. Papava, V. Olaru, and C. Sminchisescu, "Human3.6m: Large scale datasets and predictive methods for 3d human sensing in natural environments," *IEEE transactions on pattern analysis and machine intelligence*, vol. 36, no. 7, pp. 1325–1339, 2013.
- [8] M. Andriluka, L. Pishchulin, P. Gehler, and B. Schiele, "2d human pose estimation: New benchmark and state of the art analysis," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2014, pp. 3686–3693.
- [9] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, "Microsoft coco: Common objects in context," in *European conference on computer vision*. Springer, 2014, pp. 740–755.
- [10] H. Joo, H. Liu, L. Tan, L. Gui, B. Nabbe, I. Matthews, T. Kanade, S. Nobuhara, and Y. Sheikh, "Panoptic studio: A massively multiview system for social motion capture," in *Proceedings of the IEEE international conference on computer vision*, 2015, pp. 3334–3342.
- [11] T. Simon, H. Joo, I. Matthews, and Y. Sheikh, "Hand keypoint detection in single images using multiview bootstrapping," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 1145–1153.
- [12] H. Joo, T. Simon, X. Li, H. Liu, L. Tan, L. Gui, S. Banerjee, T. Godisart, B. Nabbe, I. Matthews et al., "Panoptic studio: A massively multiview system for social interaction capture," *IEEE transactions on pattern analysis and machine intelligence*, vol. 41, no. 1, pp. 190–204, 2017.
- [13] S. Jin, L. Xu, J. Xu, C. Wang, W. Liu, C. Qian, W. Ouyang, and P. Luo, "Whole-body human pose estimation in the wild," in *European Conference on Computer Vision*. Springer, 2020, pp. 196–214.
- [14] L. Xu, S. Jin, W. Liu, C. Qian, W. Ouyang, P. Luo, and X. Wang, "Zoomnas: searching for whole-body human pose estimation in the wild," *IEEE transactions on pattern analysis and machine intelligence*, vol. 45, no. 4, pp. 5296–5313, 2022.
- [15] C. Zheng, W. Wu, C. Chen, T. Yang, S. Zhu, J. Shen, N. Kehtarnavaz, and M. Shah, "Deep learning-based human pose estimation: survey," *CM computing surveys*, vol. 56, no. 1, pp. 1–37, 2023.
- [16] T. L. Munez, Y. Z. Jembre, H. T. Weldegebril, L. Chen, C. Huang, and C. Yang, "The progress of human pose estimation: survey and taxonomy of models applied in 2d human pose estimation," *IEEE access*, vol. 8, pp. 133 330–133 348, 2020.
- [17] T. B. Moeslund, A. Hilton, and V. Krüger, "A survey of advances in vision-based human motion capture and analysis," *Computer vision and image understanding*, vol. 104, no. 2-3, pp. 90–126, 2006.
- [18] G. V. Kale and V. H. Patil, "A study of vision based human motion recognition and analysis," *International journal of ambient computing and intelligence*, vol. 7, no. 2, pp. 75–92, 2016.
- [19] E. Van der Kruk and M. M. Reijne, "Accuracy of human motion capture systems for sport applications: state-of-the-art review," *European journal of sport science*, vol. 18, no. 6, pp. 806–819, 2018.
- [20] Z. Zhang, "Microsoft kinect sensor and its effect," *IEEE multimedia*, vol. 19, no. 2, pp. 4–10, 2012.
- [21] S. Springer and G. Yögev Seligmann, "Validity of the kinect for gait assessment: a focused review," *Sensors*, vol. 16, no. 2, p. 194, 2016.
- [22] "Vicon motion capture systems," <https://www.vicon.com>, accessed: 2023-08-02.
- [23] "Qualisys motion capture systems," <https://www.qualisys.com>, accessed: 2023-08-02.
- [24] M. Ota, H. Tateuchi, T. Hashiguchi, T. Kato, Y. Ogino, M. Yamagata, and N. Ichihashi, "Verification of reliability and validity of motion analysis systems during bilateral squat using human pose tracking algorithm," *Gait & posture*, vol. 80, pp. 62–67, 2020.
- [25] M. Ota, H. Tateuchi, T. Hashiguchi, and N. Ichihashi, "Verification of validity of gait analysis systems during treadmill walking and running using human pose tracking algorithm," *Gait & posture*, vol. 85, pp. 290–297, 2021.
- [26] L. VanKeersbilck, I. Hedengren, M. Clark, and I. Hunter, "Joint angle calculations using motion capture and deep learning pose estimation while running," in *International journal of exercise science: conference proceedings*, vol. 14, no. 2, 2022, p. 179.
- [27] J. Docekal, J. Rozlavec, J. Matas, and M. Hoffmann, "Human keypoint detection for close proximity human-robot interaction," in *IEEE-RAS 21st international conference on humanoid robots (humanoids)*. IEEE, 2022, pp. 450–457.
- [28] T. B. d. Lafayette, V. H. d. L. Kunst, P. V. d. S. Melo, P. d. O. Guedes, J. M. X. N. Teixeira, C. R. d. Vasconcelos, V. Teichrieb, and E. F. da Gama, "Validation of angle estimation based on body tracking data from rgb-d and rgb cameras for biomechanical assessment," *Sensors*, vol. 23, no. 1, p. 3, 2022.
- [29] C. S. T. Hii, K. B. Gan, N. Zainal, N. Mohamed Ibrahim, S. Azmin, S. H. Mat Desa, B. van de Warrenburg, and H. W. You, "Automated gait analysis based on a marker-free pose estimation model," *Sensors*, vol. 23, no. 14, p. 6489, 2023.
- [30] I. Sáradi, T. Linder, K. O. Arras, and B. Leibe, "Metraabs: metric-scale truncation-robust heatmaps for absolute 3d human pose estimation," *IEEE transactions on biometrics, behavior, and identity science*, vol. 3, no. 1, pp. 16–30, 2020.
- [31] W. Li, H. Liu, H. Tang, P. Wang, and L. Van Gool, "Mhformer: Multi-hypothesis transformer for 3d human pose estimation," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 13 147–13 156.
- [32] Z. Cao, G. Hidalgo, T. Simon, S.-E. Wei, and Y. Sheikh, "Openpose: Realtime multi-person 2d pose estimation using part affinity fields," *IEEE transactions on pattern analysis and machine intelligence*, vol. 43, no. 1, pp. 172–186, 2019.
- [33] H.-S. Fang, J. Li, H. Tang, C. Xu, H. Zhu, Y. Xiu, Y.-L. Li, and C. Lu, "Lphapose: Whole-body regional multi-person pose estimation and tracking in real-time," *IEEE transactions on pattern analysis and machine intelligence*, vol. 45, no. 6, pp. 7157–7173, 2022.
- [34] N. Sarafianos, B. Boteanu, B. Ionescu, and I. Kakadiaris, "3d human pose estimation: A review of the literature and analysis of covariates," *Computer vision and image understanding*, vol. 152, pp. 1–20, 2016.
- [35] T. J. Lopes, D. Ferrari, J. Ioannidis, M. Simic, F. Mícolis de Zevedo, and E. Pappas, "Reliability and validity of frontal plane kinematics of the trunk and lower extremity measured with 2-dimensional cameras during athletic tasks: a systematic review with meta-analysis," *Journal of orthopaedic & sports physical therapy*, vol. 48, no. 10, pp. 812–822, 2018.
- [36] G. Leporace, L. Metsavaht, F. F. Gonzalez, F. de Jesus, M. Machado, E. C. Guadagnin, and M. Gomes-Neto, "Validity and reliability of two-dimensional video-based assessment to measure joint angles during running: a systematic review and meta-analysis," *Journal of biomechanics*, vol. 157, p. 111747, 2023.
- [37] L. M. Reimer, M. Kapsecker, T. Fukushima, and S. M. Jonas, "Evaluating 3d human motion capture on mobile devices," *Applied sciences*, vol. 12, no. 10, p. 4806, 2022.
- [38] Y. Wu, A. Kirillov, F. Massa, W.-Y. Lo, and R. Girshick, "Detectron2," <https://github.com/facebookresearch/detectron2>, 2019.
- [39] Q. Zhao, C. Zheng, M. Liu, P. Wang, and C. Chen, "Poseformerv2: Exploring frequency domain for efficient and robust 3d human pose

- estimation,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2023, pp. 8877–8886.
- [40] V. Bazarevsky, I. Grishchenko, K. Raveendran, T. Zhu, F. Zhang, and M. Grundmann, “Blazepose: On-device real-time body pose tracking,” *arXiv preprint arXiv:2006.10204*, 2020.
- [41] C.-Y. Wang, . Bochkovskiy, and H.-Y. M. Liao, “Yolov7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2023, pp. 7464–7475.
- [42] Z. Cao, T. Simon, S.-E. Wei, and Y. Sheikh, “Realtime multi-person 2d pose estimation using part affinity fields,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 7291–7299.
- [43] M. Li, Z. Zhou, J. Li, and X. Liu, “Bottom-up pose estimation of multiple person with bounding box constraint,” in *24th international conference on pattern recognition*. IEEE, 2018, pp. 115–120.
- [44] B. Cheng, B. Xiao, J. Wang, H. Shi, T. S. Huang, and L. Zhang, “Higherhrnet: Scale-aware representation learning for bottom-up human pose estimation,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 5386–5395.
- [45] Y. Cheng, B. Wang, B. Yang, and R. T. Tan, “Monocular 3d multi-person pose estimation by integrating top-down and bottom-up networks,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2021, pp. 7649–7659.
- [46] J. Martinez, R. Hossain, J. Romero, and J. J. Little, “simple yet effective baseline for 3d human pose estimation,” in *Proceedings of the IEEE international conference on computer vision*, 2017, pp. 2640–2649.
- [47] H.-S. Fang, Y. Xu, W. Wang, X. Liu, and S.-C. Zhu, “Learning pose grammar to encode human body configuration for 3d pose estimation,” in *Proceedings of the I conference on artificial intelligence*, vol. 32, no. 1, 2018.
- [48] W. Li, H. Liu, R. Ding, M. Liu, P. Wang, and W. Yang, “Exploiting temporal contexts with strided transformer for 3d human pose estimation,” *IEEE transactions on multimedia*, vol. 25, pp. 1282–1293, 2022.
- [49] P. Merriault, Y. Dupuis, R. Bouteau, P. Vasseur, and X. Savatier, “study of vicon system positioning performance,” *Sensors*, vol. 17, no. 7, p. 1591, 2017.
- [50] P. E. Shrout and J. L. Fleiss, “Intraclass correlations: uses in assessing rater reliability,” *Psychological bulletin*, vol. 86, no. 2, p. 420, 1979.
- [51] . S. Murugan, G. Noh, H. Jung, E. Kim, K. Kim, H. You, and B. Boufama, “Optimising computer vision-based ergonomic assessments: sensitivity to camera position and monocular 3d pose model,” *Ergonomics*, vol. 68, no. 1, pp. 120–137, 2025.
- [52] S. Mohammed, H. Siebers, and T. Preuß, “Strengths and weaknesses of 3d pose estimation and inertial motion capture system for movement therapy,” *arXiv preprint arXiv:2306.06117*, 2023.
- [53] L. Ceriola, I. Miletì, M. Donati, and F. Patanè, “Comparison of video-based algorithms for 2d human kinematics estimation: preliminary study,” in *Journal of physics: Conference series*, vol. 2590, no. 1. IOP publishing, 2023, p. 012002.
- [54] C. Wang, F. Zhang, and S. S. Ge, “comprehensive survey on 2d multi-person pose estimation methods,” *Engineering applications of artificial intelligence*, vol. 102, p. 104260, 2021.
- [55] . Sabo, N. Mittal, . Deshpande, H. Clarke, and B. Taati, “utomated, vision-based goniometry and range of motion calculation in individuals with suspected ehlers-danlos syndromes/generalized hypermobility spectrum disorders: a comparison of pose-estimation libraries to goniometric measurements,” *IEEE journal of translational engineering in health and medicine*, vol. 12, pp. 140–150, 2023.
- [56] W. Li, R. Passama, V. Bonnet, and . Cherubini, “comparison of human skeleton extractors for real-time human-robot interaction,” in *IEEE international conference on advanced robotics and its social impacts*. IEEE, 2023, pp. 159–165.
- [57] S. Kreiss, L. Bertoni, and . lahi, “Openpipaf: Composite fields for semantic keypoint detection and spatio-temporal association,” *IEEE transactions on intelligent transportation systems*, vol. 23, no. 8, pp. 13 498–13 511, 2021.
- [58] “rkit: Capturing body motion in 3d,” https://developer.apple.com/documentation/arkit/arkit_in_ios/content_anchors/capturing_body_motion_in_3d, accessed: 2023-08-31.
- [59] C. R. Harris, K. J. Millman, S. J. Van Der Walt, R. Gommers, P. Virtanen, D. Cournapeau, E. Wieser, J. Taylor, S. Berg, N. J. Smith *et al.*, “rray programming with numpy,” *nature*, vol. 585, no. 7825, pp. 357–362, 2020.
- [60] R. Vallat, “Pingouin: statistics in python,” *J. open source Softw.*, vol. 3, no. 31, p. 1026, 2018.
- [61] R. Vallat, “intraclass_corr function in pingouin,” accessed: 2025-03-25. [Online]. available: https://pingouin-stats.org/build/html/_modules/pingouin/reliability.html#intraclass_corr
- [62] B. L. Welch, “On the comparison of several mean values: an alternative approach,” *Biometrika*, vol. 38, no. 3/4, pp. 330–336, 1951.
- [63] M. Delacre, D. Lakens, and C. Leys, “Why psychologists should by default use welch’s t-test instead of student’s t-test,” *International review of social psychology*, vol. 30, no. 1, pp. 92–101, 2017.
- [64] H. Liu, *Comparing Welch NOV , a Kruskal-Wallis test, and traditional NOV in case of heterogeneity of variance*. Virginia commonwealth university, 2015.
- [65] R. Vallat, “welch_anova function in pingouin,” accessed: 2025-03-25. [Online]. available: https://pingouin-stats.org/build/html/_modules/pingouin/parametric.html#welch_anova
- [66] R. Vallat, “ttesti function in pingouin,” accessed: 2025-03-25. [Online]. available: https://pingouin-stats.org/build/html/_modules/pingouin/parametric.html#ttesti
- [67] Y. Cheng, B. Wang, and R. T. Tan, “Dual networks based 3d multi-person pose estimation from monocular video,” *IEEE transactions on pattern analysis and machine intelligence*, vol. 45, no. 2, pp. 1636–1651, 2022.
- [68] G. Moon, J. Y. Chang, and K. M. Lee, “Camera distance-aware top-down approach for 3d multi-person pose estimation from a single rgb image,” in *Proceedings of the IEEE/CVF international conference on computer vision*, 2019, pp. 10 133–10 142.
- [69] H.-S. Fang, S. Xie, Y.-W. Tai, and C. Lu, “Rmpe: Regional multi-person pose estimation,” in *Proceedings of the IEEE international conference on computer vision*, 2017, pp. 2334–2343.
- [70] J. Li, C. Wang, H. Zhu, Y. Mao, H.-S. Fang, and C. Lu, “Crowdpose: Efficient crowded scenes pose estimation and a new benchmark,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 10 863–10 872.
- [71] T. Chen, C. Fang, X. Shen, Y. Zhu, Z. Chen, and J. Luo, “natomy-aware 3d human pose estimation with bone-based pose decomposition,” *IEEE transactions on circuits and systems for video technology*, vol. 32, no. 1, pp. 198–209, 2021.
- [72] D. Pavllo, C. Feichtenhofer, D. Grangier, and M. uli, “3d human pose estimation in video with temporal convolutions and semi-supervised training,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 7753–7762.
- [73] R. Liu, J. Shen, H. Wang, C. Chen, S.-c. Cheung, and V. sari, “tention mechanism exploits temporal contexts: Real-time 3d human pose reconstruction,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 5064–5073.
- [74] D. Wang and S. Zhang, “Contextual instance decoupling for robust multi-person pose estimation,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 11 060–11 068.
- [75] Q. Zhao, C. Zheng, M. Liu, and C. Chen, “single 2d pose with context is worth hundreds for 3d human pose estimation,” *advances in neural information processing systems*, vol. 36, pp. 27 394–27 413, 2023.
- [76] X. Ma, J. Su, C. Wang, H. Ci, and Y. Wang, “Context modeling in 3d human pose estimation: unified perspective,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2021, pp. 6238–6247.
- [77] C. Zheng, S. Zhu, M. Mendieta, T. Yang, C. Chen, and Z. Ding, “3d human pose estimation with spatial and temporal transformers,” in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 11 656–11 665.
- [78] K. Isakov, E. Burkov, V. Lempitsky, and Y. Malkov, “Learnable triangulation of human pose,” in *Proceedings of the IEEE/CVF international conference on computer vision*, 2019, pp. 7718–7727.
- [79] H. Wang, L. Zhou, Y. Chen, M. Tang, and J. Wang, “Regularizing vector embedding in bottom-up human pose estimation,” in *European conference on computer vision*. Springer, 2022, pp. 107–122.
- [80] F. Zhang, X. Zhu, H. Dai, M. Ye, and C. Zhu, “Distribution-aware coordinate representation for human pose estimation,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 7093–7102.
- [81] Z. Wang, X. Nie, X. Qu, Y. Chen, and S. Liu, “Distribution-aware single-stage models for multi-person 3d pose estimation,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 13 096–13 105.
- [82] Z. Liu, H. Chen, R. Feng, S. Wu, S. Ji, B. Yang, and X. Wang, “Deep dual consecutive network for human pose estimation,” in *Proceedings*

- of the *IEEE/CVF conference on computer vision and pattern recognition*, 2021, pp. 525–534.
- [83] K. Sun, B. Xiao, D. Liu, and J. Wang, “Deep high-resolution representation learning for human pose estimation,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 5693–5703.
- [84] B. Xiao, H. Wu, and Y. Wei, “Simple baselines for human pose estimation and tracking,” in *Proceedings of the European conference on computer vision*, 2018, pp. 466–481.
- [85] J. Wang, K. Sun, T. Cheng, B. Jiang, C. Deng, Y. Zhao, D. Liu, Y. Mu, M. Tan, X. Wang *et al.*, “Deep high-resolution representation learning for visual recognition,” *IEEE transactions on pattern analysis and machine intelligence*, vol. 43, no. 10, pp. 3349–3364, 2020.
- [86] Z. Geng, K. Sun, B. Xiao, Z. Zhang, and J. Wang, “Bottom-up human pose estimation via disentangled keypoint regression,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2021, pp. 14 676–14 686.
- [87] M. Véges and . Lőrincz, “Absolute human pose estimation with depth prediction network,” in *International joint conference on neural networks*. IEEE, 2019, pp. 1–7.
- [88] J. Yang, . Zeng, S. Liu, F. Li, R. Zhang, and L. Zhang, “Explicit box detection unifies end-to-end multi-person pose estimation,” *arXiv preprint arXiv:2302.01593*, 2023.
- [89] Y. Cai, L. Ge, J. Liu, J. Cai, T.-J. Cham, J. Yuan, and N. M. Thalmann, “Exploiting spatial-temporal relationships for 3d pose estimation via graph convolutional networks,” in *Proceedings of the IEEE/CVF international conference on computer vision*, 2019, pp. 2272–2281.
- [90] S. Yan, Y. Xiong, and D. Lin, “Spatial temporal graph convolutional networks for skeleton-based action recognition,” in *Proceedings of the I conference on artificial intelligence*, vol. 32, no. 1, 2018.
- [91] X. Wang, R. Girshick, . Gupta, and K. He, “Non-local neural networks,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 7794–7803.
- [92] Z. Liu, R. Feng, H. Chen, S. Wu, Y. Gao, Y. Gao, and X. Wang, “Temporal feature alignment and mutual information maximization for video-based human pose estimation,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 11 006–11 016.
- [93] C. Li and G. H. Lee, “Generating multiple hypotheses for 3d human pose estimation with mixture density network,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 9887–9895.
- [94] S. Sharma, P. T. Varigonda, P. Bindal, . Sharma, and . Jain, “Monocular 3d human pose estimation by generation and ordinal ranking,” in *Proceedings of the IEEE/CVF international conference on computer vision*, 2019, pp. 2325–2334.
- [95] T. He, Z. Zhang, H. Zhang, Z. Zhang, J. Xie, and M. Li, “Bag of tricks for image classification with convolutional neural networks,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 558–567.
- [96] Y. Cheng, B. Wang, B. Yang, and R. T. Tan, “Graph and temporal convolutional networks for 3d multi-person pose estimation in monocular videos,” in *Proceedings of the I conference on artificial intelligence*, vol. 35, no. 2, 2021, pp. 1157–1165.
- [97] Y.-L. Li, L. Xu, X. Liu, X. Huang, Y. Xu, S. Wang, H.-S. Fang, Z. Ma, M. Chen, and C. Lu, “Pastanet: Toward human activity knowledge engine,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 382–391.
- [98] K. Zhou, X. Han, N. Jiang, K. Jia, and J. Lu, “Hemlets pose: Learning part-centric heatmap triplets for accurate 3d human pose estimation,” in *Proceedings of the IEEE/CVF international conference on computer vision*, 2019, pp. 2344–2353.
- [99] K. Zhou, X. Han, N. Jiang, K. Jia, and J. Lu, “Hemlets posh: learning part-centric heatmap triplets for 3d human pose and shape estimation,” *IEEE transactions on pattern analysis and machine intelligence*, vol. 44, no. 6, pp. 3000–3014, 2021.
- [100] B. Cheng, B. Xiao, J. Wang, H. Shi, T. S. Huang, and L. Zhang, “Higherhmet: Scale-aware representation learning for bottom-up human pose estimation,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 5386–5395.
- [101] K. Sun, B. Xiao, D. Liu, and J. Wang, “Deep high-resolution representation learning for human pose estimation,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 5693–5703.
- [102] J. Wang, K. Sun, T. Cheng, B. Jiang, C. Deng, Y. Zhao, D. Liu, Y. Mu, M. Tan, X. Wang *et al.*, “Deep high-resolution representation learning for visual recognition,” *IEEE transactions on pattern analysis and machine intelligence*, vol. 43, no. 10, pp. 3349–3364, 2020.
- [103] Y. Yuan, R. Fu, L. Huang, W. Lin, C. Zhang, X. Chen, and J. Wang, “Hrformer: high-resolution transformer for dense prediction,” in *Proceedings of the 35th international conference on neural information processing systems*, 2021, pp. 7281–7293.
- [104] J. Lin and G. H. Lee, “Hdnet: Human depth estimation for multi-person camera-space localization,” in *European conference on computer vision*. Springer, 2020, pp. 633–648.
- [105] J. Li, W. Su, and Z. Wang, “Simple pose: Rethinking and improving a bottom-up approach for multi-person pose estimation,” in *Proceedings of the I conference on artificial intelligence*, vol. 34, no. 07, 2020, pp. 11 354–11 361.
- [106] Y. Zhang, Y. Wang, O. Camps, and M. Szaier, “Key frame proposal network for efficient pose estimation in videos,” in *European conference on computer vision*. Springer, 2020, pp. 609–625.
- [107] G. Rogez, P. Weinzaepfel, and C. Schmid, “Lcr-net: Localization-classification-regression for human pose,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 3433–3441.
- [108] G. Rogez, P. Weinzaepfel, and C. Schmid, “Lcr-net++: Multi-person 2d and 3d pose detection in natural images,” *IEEE transactions on pattern analysis and machine intelligence*, vol. 42, no. 5, pp. 1146–1161, 2019.
- [109] G. Ning, J. Pei, and H. Huang, “Lighttrack: generic framework for online top-down human pose tracking,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition workshops*, 2020, pp. 1034–1035.
- [110] C. Yu, B. Xiao, C. Gao, L. Yuan, L. Zhang, N. Sang, and J. Wang, “Lite-hrnet: lightweight high-resolution network,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2021, pp. 10 440–10 450.
- [111] Y. Wang, M. Li, H. Cai, W.-M. Chen, and S. Han, “Lite pose: Efficient architecture design for 2d human pose estimation,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 13 126–13 136.
- [112] M. Fabbri, F. Lanzi, S. Calderara, S. Ietto, and R. Cucchiara, “Compressed volumetric heatmaps for multi-person 3d pose estimation,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 7204–7213.
- [113] “Mediapipe solution guide,” <https://developers.google.com/mediapipe/solutions/guide>, accessed: 2023-08-07.
- [114] “Mediapipe pose landmark detection guide,” https://developers.google.com/mediapipe/solutions/vision/pose_landmarker, accessed: 2023-08-07.
- [115] I. Sárándi, . Hermans, and B. Leibe, “Learning 3d human pose estimation from dozens of datasets using a geometry-aware autoencoder to bridge between skeleton formats,” in *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, 2023, pp. 2956–2966.
- [116] W. Li, H. Liu, H. Tang, and P. Wang, “Multi-hypothesis representation learning for transformer-based 3d human pose estimation,” *Pattern recognition*, vol. 141, p. 109631, 2023.
- [117] M. Contributors, “Openmmlab pose estimation toolbox and benchmark,” <https://github.com/open-mmlab/mmpose>, 2020.
- [118] Z. Zou and W. Tang, “Modulated graph convolutional network for 3d human pose estimation,” in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 11 477–11 487.
- [119] L. Zhao, X. Peng, Y. Tian, M. Kapadia, and D. N. Metaxas, “Semantic graph convolutional networks for 3d human pose regression,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 3425–3435.
- [120] W. Zhu, X. Ma, Z. Liu, L. Liu, W. Wu, and Y. Wang, “Motionbert: unified perspective on learning human motion representations,” in *Proceedings of the IEEE/CVF international conference on computer vision*, 2023, pp. 15 085–15 099.
- [121] “Movenet: Ultra fast and accurate pose detection model,” <https://www.tensorflow.org/hub/tutorials/movenet>, accessed: 2024-08-21.
- [122] W. Li, Z. Wang, B. Yin, Q. Peng, Y. Du, T. Xiao, G. Yu, H. Lu, Y. Wei, and J. Sun, “Rethinking on multi-stage networks for human pose estimation,” *arXiv preprint arXiv:1901.00148*, 2019.
- [123] Y. Chen, Z. Wang, Y. Peng, Z. Zhang, G. Yu, and J. Sun, “Cascaded pyramid network for multi-person pose estimation,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 7103–7112.
- [124] S. Kreiss, L. Bertoni, and . lahi, “Pifpaf: Composite fields for human pose estimation,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 11 977–11 986.

- [125] T. Simon, H. Joo, I. Matthews, and Y. Sheikh, "Hand keypoint detection in single images using multiview bootstrapping," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 1145–1153.
- [126] S.-E. Wei, V. Ramakrishna, T. Kanade, and Y. Sheikh, "Convolutional pose machines," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 4724–4732.
- [127] L. Li, J. Miao, D. Shi, W. Tan, Y. Ren, Y. Yang, and S. Pu, "Distilling detr with visual-linguistic knowledge for open-vocabulary object detection," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2023, pp. 6501–6510.
- [128] D. Shi, X. Wei, L. Li, Y. Ren, and W. Tan, "End-to-end multi-person pose estimation with transformers," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 11 069–11 078.
- [129] X. Yu, D. Shi, X. Wei, Y. Ren, T. Ye, and W. Tan, "Soit: Segmenting objects with instance-aware transformers," in *Proceedings of the I conference on artificial intelligence*, vol. 36, no. 3, 2022, pp. 3188–3196.
- [130] D. Shi, X. Wei, X. Yu, W. Tan, Y. Ren, and S. Pu, "Inspose: instance-aware networks for single-stage multi-person pose estimation," in *Proceedings of the 29th CM international conference on multimedia*, 2021, pp. 3079–3087.
- [131] P. authors, "Paddledetection, object detection and instance segmentation toolkit based on paddlepaddle," <https://github.com/PaddlePaddle/PaddleDetection>, 2019.
- [132] P. Panteleris and . rgyros, "Pe-former: Pose estimation transformer," in *International conference on pattern recognition and artificial intelligence*. Springer, 2022, pp. 3–14.
- [133] J. N. Kundu, S. Seth, V. Jampani, M. Rakesh, R. V. Babu, and . Chakraborty, "Self-supervised 3d human pose estimation via part guided novel image synthesis," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 6152–6162.
- [134] F. Wei, X. Sun, H. Li, J. Wang, and S. Lin, "Point-set anchors for object detection, instance segmentation and pose estimation," in *European conference on computer vision*. Springer, 2020, pp. 527–544.
- [135] M. Véges and . Lőrincz, "Temporal smoothing for 3d human pose estimation and localization for occluded people," in *International conference on neural information processing*. Springer, 2020, pp. 557–568.
- [136] D. C. Luvizon, H. Tabia, and D. Picard, "Human pose regression by combining indirect part detection and contextual information," *Computers & graphics*, vol. 85, pp. 15–22, 2019.
- [137] W. Mao, Y. Ge, C. Shen, Z. Tian, X. Wang, Z. Wang, and . v. den Hengel, "Poseur: Direct human pose regression with transformers," in *European conference on computer vision*. Springer, 2022, pp. 72–88.
- [138] G. Bertasius, C. Feichtenhofer, D. Tran, J. Shi, and L. Torresani, "Learning temporal pose estimation from sparsely-labeled videos," *advances in neural information processing systems*, vol. 32, 2019.
- [139] H. Ma, Z. Wang, Y. Chen, D. Kong, L. Chen, X. Liu, X. Yan, H. Tang, and X. Xie, "Ppt: token-pruned pose transformer for monocular and multi-view human pose estimation," in *European conference on computer vision*. Springer, 2022, pp. 424–442.
- [140] K. Li, S. Wang, X. Zhang, Y. Xu, W. Xu, and Z. Tu, "Pose recognition with cascade transformers," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2021, pp. 1944–1953.
- [141] H. Liu, F. Liu, X. Fan, and D. Huang, "Polarized self-attention: Towards high-quality pixel-wise regression," *arXiv preprint arXiv:2107.00782*, 2021.
- [142] B. Wandt and B. Rosenhahn, "Repnet: Weakly supervised training of an adversarial reprojection network for 3d human pose estimation," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 7782–7791.
- [143] J. Li, S. Bian, . Zeng, C. Wang, B. Pang, W. Liu, and C. Lu, "Human pose regression with residual log-likelihood estimation," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 11 025–11 034.
- [144] Y. Cai, Z. Wang, B. Yin, R. Yin, . Du, Z. Luo, Z. Li, X. Zhou, G. Yu, E. Zhou *et al.*, "Res-steps-net for multi-person pose estimation," in *Joint COCO and Mapillary workshop at ICCV*, 2019, p. 3.
- [145] Y. Cai, Z. Wang, Z. Luo, B. Yin, . Du, H. Wang, X. Zhang, X. Zhou, E. Zhou, and J. Sun, "Learning delicate local representations for multi-person pose estimation," in *European conference on computer vision*. Springer, 2020, pp. 455–472.
- [146] T. Jiang, "rtmllib," <https://github.com/Tau-J/rtmllib>, 2023.
- [147] T. Jiang, P. Lu, L. Zhang, N. Ma, R. Han, C. Lyu, Y. Li, and K. Chen, "RtmPose: Real-time multi-person pose estimation based on mmpose," *arXiv preprint arXiv:2303.07399*, 2023.
- [148] P. Lu, T. Jiang, Y. Li, X. Li, K. Chen, and W. Yang, "Rtmo: Towards high-performance one-stage real-time multi-person pose estimation," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2024, pp. 1491–1500.
- [149] T. Jiang, X. Xie, and Y. Li, "Rtmw: Real-time multi-person 2d and 3d whole-body pose estimation," *arXiv preprint arXiv:2407.08634*, 2024.
- [150] J. Zhen, Q. Fang, J. Sun, W. Liu, W. Jiang, H. Bao, and X. Zhou, "Smap: Single-shot multi-person absolute 3d pose estimation," in *European conference on computer vision*. Springer, 2020, pp. 550–566.
- [151] . Zeng, X. Sun, F. Huang, M. Liu, Q. Xu, and S. Lin, "Srnet: Improving generalization in 3d human pose estimation with a split-and-recombine approach," in *European conference on computer vision*. Springer, 2020, pp. 507–523.
- [152] Z. Luo, Z. Wang, Y. Huang, L. Wang, T. Tan, and E. Zhou, "Rethinking the heatmap regression for bottom-up human pose estimation," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2021, pp. 13 264–13 273.
- [153] Y. Li, S. Zhang, Z. Wang, S. Yang, W. Yang, S.-T. Xia, and E. Zhou, "Tokenpose: Learning keypoint tokens for human pose estimation," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 11 313–11 322.
- [154] S. Yang, Z. Quan, M. Nie, and W. Yang, "Transpose: Keypoint localization via transformer," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 11 802–11 812.
- [155] Y. Li, M. Hao, Z. Di, N. B. Gundavarapu, and X. Wang, "Test-time personalization with a transformer for human pose estimation," *advances in neural information processing systems*, vol. 34, pp. 2583–2597, 2021.
- [156] J. Huang, Z. Zhu, F. Guo, and G. Huang, "The devil is in the details: Delving into unbiased data processing for human pose estimation," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 5700–5709.
- [157] J. Huang, Z. Zhu, G. Huang, and D. Du, " id: Pushing the performance boundary of human pose estimation with information dropping augmentation," *arXiv preprint arXiv:2008.07139*, 2020.
- [158] G. Jocher, . Chaurasia, and J. Qiu, "Yolo by ultralytics," jan 2023, accessed: 2023-11-23. [Online]. available: <https://github.com/ultralytics/ultralytics>
- [159] B. rtacho and . Savakis, "Unipose: Unified human pose estimation in single images and videos," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 7035–7044.
- [160] Q. Nie, Z. Liu, and Y. Liu, "Unsupervised 3d human pose representation with viewpoint and pose disentanglement," in *European conference on computer vision*. Springer, 2020, pp. 102–118.
- [161] Y. Xu, J. Zhang, Q. Zhang, and D. Tao, "Vitpose: Simple vision transformer baselines for human pose estimation," *advances in neural information processing systems*, vol. 35, pp. 38 571–38 584, 2022.
- [162] M. Véges and . Lőrincz, "Multi-person absolute 3d human pose estimation with weak depth supervision," in *International conference on artificial neural networks*. Springer, 2020, pp. 258–270.
- [163] C.-Y. Wang, H.-Y. M. Liao, and I.-H. Yeh, "Designing network design strategies through gradient path analysis," *Journal of information science & engineering*, vol. 39, no. 4, 2023.
- [164] D. Maji, S. Nagori, M. Mathew, and D. Poddar, "Yolo-pose: Enhancing yolo for multi person pose estimation using object keypoint similarity loss," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 2637–2646.

VIII. BIOGRAPHY SECTION



Fabian Kahl is a Ph.D. candidate at the Institute for Digital Medicine, University Hospital Bonn. He holds an MSc in Computer Science from the University of Bonn, Germany, including a semester abroad at the Università della Svizzera italiana in Lugano, Switzerland. His research interests include computer vision and machine learning in medicine.



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Jennifer Faber is head of the Ataxia Clinic at the Center of Neurology, University Hospital Bonn, and clinical investigator at the German Center for Neurodegenerative Diseases (DZNE), Bonn, Germany. For the integrated concept of comprehensive patient care and excellent ataxia research, the Ataxia Clinic was awarded as the first European "Ataxia Center of Excellence" by the National Ataxia Foundation (NAF) in 2023. Dr. Jennifer Faber received a pre-diploma in Mathematics from Bonn University, and graduated with an MD and doctor of medicine degree. Her research focuses are biomarkers in ataxias, particularly those preceding the clinical onset.